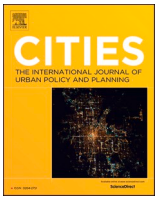
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Does digital space shape physical space? Unraveling spatial characteristics of Third Places distinguished by online platform using explainable AI

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ABSTRACT

Although it is evident that digital technology is reshaping urban spaces, the absence of a quantitative research framework has hindered the understanding of the empirical impact of digital technology on physical spaces. Through the lens of platform urbanism, this study aims to unravel the spatial characteristics of physical space as differentiated by digital technology, focusing on Third Places, where people socially interact. We quantitatively compared the spatial attributes of four distinct types of Third Places, categorized based on the extent to which they rely on digital technology to offer services, defining a Third Place as a "Digi-Third Place." We employed two data sources: business licensing data and NAVER web-crawled data. We conducted three quantitative analyses, ANOVA F-test, mean comparison, and classification, to verify the spatial differences among Digi-Third Places. The results reveal that Third Places exhibit significant differences in location and building characteristics. We quantitatively elucidated the spatial characteristics of Digi-Third Places: providing additional accessibility via an online platform; Non-Digi-Third Places: securing advantaged accessibility to sufficient visitors; and Advanced Digi-Third Places: utilizing online reservations for service competition. This study offers a valuable contribution by facilitating empirical analyses that aim to unravel the interactions between physical space and digital technology.

1. Introduction

The advent of digital technology has significantly influenced human behaviors within urban environments, encompassing domains such as quality of life (Ali et al., 2020), transportation behaviors (Ben-Elia & Zhen, 2018; Mokhtarian et al., 2006), and daily activities (Ding & Lu, 2017). The spatial influence of digital technology is even comparable to the spatial impact of cars and elevators on urban form (Afradi & Nourian, 2022). However, because of the complexity of the mechanism by which digital technology indirectly influences urban space (Afradi & Nourian, 2022), the lack of a quantitative research framework has hindered empirical research on the spatial impact of digital technology on urban space.

In the academic agenda, the field of digital geography has emerged as a prominent domain that offers valuable conceptual frameworks for investigating the intricate relationship between digital technology and physical space (Ash et al., 2018; Osborne & Jones, 2023). Prominent studies in the digital geography field have often leveraged qualitative research methodologies (Kukka et al., 2014; Lynch, 2020; McNeill, 2021). Notably, however, quantitative methodologies for investigating empirical phenomena remain scarce. Consequently, there is a pressing need to address a crucial gap in the scholarly landscape by creating innovative research frameworks that facilitate quantitative analyses of digital technology's spatial influence within urban contexts.

The concept of the Third Place, originally proposed by Oldenburg and Brissett (1982), delineates spaces where individuals can participate in social interactions away from their homes and workplaces. The advent of digital technology, such as social media services or online games, has engendered notable shifts in the modes of visiting Third Places and their accessibility (Steinkuehler & Williams, 2006; Yuen & Johnson, 2017). Therefore, Third Places that are tangible physical environments are also affected by digital technology due to the ways in which technology reshapes individuals' behaviors.

This study aims to unravel the spatial characteristics of physical urban spaces differentiated by digital technology, with a specific focus on the Third Place. In pursuit of this objective, we conducted a

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0264-2751/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

quantitative analysis comparing the spatial attributes of four distinct categories of Third Places, classified based on the extent to which they rely on digital technology to offer services. Therefore, the core research question is "What is the physical difference between Third Places based on digital technology utilization?" This research endeavors to contribute to the understanding of the interaction between digital technology and physical urban spaces, thereby augmenting existing scholarly investigations that have predominantly adopted qualitative methodologies.

2. Literature review

2.1. Cyberspace and physical space

The fact that digital technology has drastically transformed the urban environment is undeniable. Digital technology has revolutionized the way individuals traverse urban space (Mokhtarian et al., 2006; Pawlak et al., 2020), influenced the spatial distribution of firms' locations (Sohn et al., 2003), and influenced shopping behavior (Gould & Golob, 1997). Eminent scholars have forecast the spatial impact of digital technology across a spectrum of dimensions. For instance, Mitchell (1996) introduced the concept of "cities of bits" to illustrate how digital technology deeply permeates physical entities by logically linking cities with virtual software. Batty (Batty, 1993; Batty & Miller, 2000) suggested that the invisible communication structure would significantly reshape tangible space, coining the term "cyberspace." Castells (2000, 2002) noted the advent of "the space of flow," wherein the network society extends physical spaces through information and communication technologies (ICT). Recently, the evolving relationship between digital technology and empirical spatial phenomena has ascended to a prominent position on the research agenda, resulting in the burgeoning field termed "digital geography." Ash et al. (2018) mentioned this epoch as a "digital turn," contending that the rapid development of digital technology is fundamentally reshaping both research methodologies within the discipline and the definitions of space. While numerous studies have emphasized the potential of digital technology to engender a new spatial paradigm, a robust research framework to empirically investigate these phenomena is still lacking (Dadashpoor & Yousefi, 2018; Yousefi & Dadashpoor, 2020). Consequently, it is challenging to directly observe the influence of digital technologies on the form of urban space or to isolate it from other factors (Audirac, 2005), necessitating the development of a robust empirical research framework.

2.2. Platform urbanism for understanding digital phenomena in urban space

As digital platforms have emerged as a pivotal mechanism and device of urban economies and organizations, platform urbanism has become a prominent agenda for understanding the transformations in urban space brought about by digital technologies. The concept of platform urbanism identifies digital platforms to examine the empirical phenomena that emerge from digital technology. Caprotti et al. (2022) defined a platform as a socio-technical assemblage that is aggregated into a technological service. They mentioned that service providers, users, and technology firms create business and social interaction through the digital platforms. Through the interaction, digital platforms restructure the power structure and material practices of cities by involving a wide range of urban services (Bissell, 2020; Hodson et al., 2023). In addition, previous studies have suggested a critical agenda for the digital transformation of cities through platforms and algorithmic technology. Habermas (2022) argues that the digital public sphere, structured by the logic of capitalism, will reduce the possibility of different perspectives through privatization and fragmentation, exacerbating the construction of a democratic discussion. Graham and Dittus (2022) investigated that the ability to participate in the system offered by platform firms reproduces knowledge production and information inequality in online spaces, focusing on Google Maps. As such, platform urbanism defines platforms as digital arrangements that reorganize economic and social structure (Murillo et al., 2017) and focuses on platforms as key actors in transforming urban space through services. Although prominent studies in the fields of platform urbanism have explored empirical phenomena, only a few have leveraged quantitative methods. While Hatuka et al. (2021) employed empirical analysis to elucidate variations in digital practices contingent upon urban lifestyles, an imperative need for an advanced empirical research framework remains. To embark on a quantitative analysis of empirical phenomena, it is necessary to identify physical research subjects that materialize the urban space where technological advancements have influenced human behavior.

2.3. Social activities in urban space and Third Places

A thorough exploration of social activities within the context of urban space is imperative. Guo et al. (2022) proposed a three-category classification of the daily activities in urban space: necessary, spontaneous, and social activities. Among the categories, social activities have been shown to facilitate the development of social capital (Williams & Pocock, 2010), community cohesion (Hickman, 2013), and even the quality of life (Lee, 2022; Oldenburg & Brissett, 1982). Furthermore, social activities enhance urban vitality, thereby creating more livable spaces (Jacobs, 1961), and creativity induced by the interaction of diverse social classes (Florida, 2002). To analyze social activities, previous studies employed human mobility data, including travel survey data (Choi et al., 2021; Morris, 2019), mobility data (Toole et al., 2015; Moro et al., 2021), and smart card data (Wu et al., 2025). Admitting that, given the established correlation between social behavior and the physical urban environment (Hickman, 2013; Whyte, 1980), there is a compelling need to examine the physical spaces where social activities take place to better understand their dynamics in urban environments.

Third Places provide individuals with opportunities for relaxation, the exchange of local knowledge, and social interaction (Oldenburg & Brissett, 1982). Oldenburg and Brissett (1982) emphasized their role in affording individuals the chance to participate in social engagement beyond the home and workplace. Third Places include social and commercial places such as restaurants, cafes, beauty shops, and libraries that offer conversation and a playful mood. Given that Third Places are known to encourage informal connections (Yuen & Johnson, 2017) and facilitate physical gatherings (Adelfio et al., 2020), they have emerged as physical sites to materialize the empirical phenomena that occur through social behavior.

The rationale for focusing on Third Places in this study is to explore informal activities that have not been addressed in previous studies in the field of digital geography and digital transformation. Previous studies on the interaction between digital technology and urban space have focused on residential and work activities, such as population and employment distribution (Audirac, 2005; Rachmawati et al., 2015), and travel behavior, including commuting and leisure (Kwan et al., 2007; Mokhtarian et al., 2006). According to Oldenburg and Brissett (1982), formal activities that occur in the housing (the first place) and workplace (the second place) were essential to live in the city. However, Oldenburg and Brissett (1982) emphasized that the unofficial public interaction that occurred in the Third Places played an instrumental role for a healthy urban lifestyle. Focusing on the exploration of informal activities that occur in urban environments transformed by digital technology that have not been addressed in previous studies, this study employs the notion of Third Places. Thus, this study offers a flexible definition of the Third Place, characterizing it as a physical place wherein individuals engage in social activities. Rigidly defined, a Third Place is an environment conducive to conversation-based interactions with a sense of belonging (Jeffres et al., 2009; Rosenbaum, 2006). However, recent studies have taken a broader perspective, defining a Third Place more

loosely as a space for group activities with friends or a setting where individuals participate in informal lifestyles (Lee, 2022; Nguyen et al., 2019). For instance, Thompson (2018) asserted that the definition of the Third Place needed reconsideration because social connections in modern urban space consist mainly of weak ties in densely populated high-rise urban environments. Thus, this study defines the Third Place as a physical place where social activities occur, intentionally excluding social activity facilitated through virtual connections (Kukka et al., 2014; Robinson & Deshano, 2011) or social service access via delivery services (Bissell, 2020).

2.4. Digi-Third Places

This study revisits the "DigiPlace" concept introduced by Zook and Graham (2007a,b), who aimed to illustrate empirical phenomena stemming from the spatial impact of digital technology on urban spaces. DigiPlace, as Zook and Graham noted, referred to query results generated by Google's ranking system having the potential to influence visitors or consumers, thereby exerting spatial effects. DigiPlace provides a considerable concept to pursue the purpose of this study, which aims to discover the spatial characteristics related to digital technology in terms of the hierarchy of urban space restructured by platforms. McNeill (2021) and Bissell (2020) reinforced the DigiPlace idea, highlighting how big-tech platforms ultimately reshape urban geography by altering consumer behaviors and economic structure. Platforms provide the opportunity to engage in social activities through a digitally-mediated environment (Thiel, 2023), and their use is an important indicator to investigate the participation in social activities that occurs in the physical environment through the medium of digital technologies. For example, Park and Kim (2024) revealed that restaurants offering online reservations through platforms have distinct spatial characteristics to existing restaurants in terms of location theory. In exploring the transformation of physical presence by digital technology, platforms play a role in mediating the interaction between service providers and users. Therefore, through the lens of platform urbanism, we can investigate the spatial characteristics of physical space that exploit digital technologies to present the opportunity to participate in social activities.

This study stratifies the level of digital utility within Third Places based on their reliance on digital technology through online platforms. First, the Non-Digi-Third Place (NDTP) group is not searchable on online platforms and exclusively offers services and information in physical form. Second, the Digi-Third Place (DTP) group encompasses venues that are accessible and searchable on online platforms, thus disseminating information via digital technology. Last, the Advanced Digi-Third Place (ADTP) group represents places that provide online reservation services, including discounts and seat reservations, through online platforms, affording platform users priority and advantages through the utilization of digital technology. These three groups serve as tangible materializations of empirical phenomena, showcasing how physical spaces leverage different levels of digital technology to offer individuals opportunities for social interaction.

3. Materials and methods

This study aims to investigate the spatial difference in physical space based on varying levels of digital technology through Third Places-physical space where individuals gather and participate in social interaction. To pursue the research objective with a quantitative research framework, we set three specific research subquestions: (1) Does the extent of digital utility result in discernible variations in the spatial attributes of Third Places? (2) What spatial differences are observed among Third Places based on the degree of digital technology utility? (3) What are the critical determinants of these identified spatial characteristics?

To evaluate and interpret the results of the quantitative analysis empirically, we employed geospatial analysis for specific locations at the research sites. To pursue these aims, we set our research process (Fig. 1).

3.1. Study area

For this study, we selected Seoul, the capital of South Korea, which encompasses a land area of approximately 605.2 km^2 and has a population of approximately 9.5 million. Seoul is one of the most exemplary global smart cities, rendering it a compelling site for analyzing the vibrant interplay between digital technology and urban spatial dynamics. Seoul boasts an extensive and pervasive adoption of ICT, with minimal evidence of a digital divide and widespread access to broadband connections (Park et al., 2015; Southern & Townsend, 2005; Wellman & Chen, 2004). For example, 1) 99.7 % of domestic households and 96.49 % of individuals are able to access the internet (OECD, 2021), and 2) 95 % of citizens in Seoul possess a smartphone (Pew Research Center, 2019). These figures unequivocally attest to the profound utilization of digital technology, including online platforms and reservation services, for individuals' mundane activities.

3.2. Dataset

3.2.1. Categories in Third Places

Third Places encompass a variety of places, each characterized by distinct activities that take place within it. The locations and visiting patterns of Third Places vary in accordance with the types of activities that occur within them. To systematically address this variety, we divided Third Places in four categories, each distinguished by the predominant activities that occurred within it. These categories, namely, "Eating, Drinking, and Talking" (hereafter Eating), "Retail and Local" (hereafter Retail), "Club and Organization" (hereafter Club), and "Education & Recreation" (hereafter Education), were informed by previous studies (Jeffres et al., 2009; Lee, 2022) and are visually depicted in Fig. 2.

Notably, within the Education category, we considered the unique local context of Seoul, where individuals frequently take advantage of private tutoring services or facilities (*hagwon*) for self-improvement and amusement. Consequently, the categorization of Third Places included educational and recreational institutions within one category. Furthermore, the classification of Third Places according to the level of digital utility resulted in the identification of three distinct groups, as illustrated in Fig. 3.

3.2.2. Data acquisition and spatial database

To acquire DTP and ADTP data, we employed a web crawling method to extract queried data from NAVER, South Korea's most prominent website and search engine. NAVER operates NAVER Place, a specialized location-based platform that provides navigation and rating services to users. Self-employed business owners seeking to advertise their enterprises register them on NAVER Place. Consumers are able to approach location-based services, such as the navigation, menu information, reviews, ratings, and reservation services of a restaurant or beauty shop, without visiting a physical venue. Given NAVER's extensive database of location-based information and commercial services tailored to supporting self-employed business owners, it was an ideal data source for this research purpose. From the perspective of platform urbanism, registration on NAVER place has two remarkable implications. 1) It provides a place of advertisement to expand their business and increase their attractiveness for providers and to explore options for reasonable consumption for users (Caprotti et al., 2022). 2) It participates in the production of information led by technology firms by accumulating review and rating data (Zook & Graham, 2007a,b). Among the complex nature of digital interaction between cyberspace and physical space, this study focuses on discovering the spatial characteristics of Third Places that are distinguished by the use of platforms.

To gather DTP data, we collected all queried data from NAVER by searching the keywords "road name" + "place type." Here, "road name"

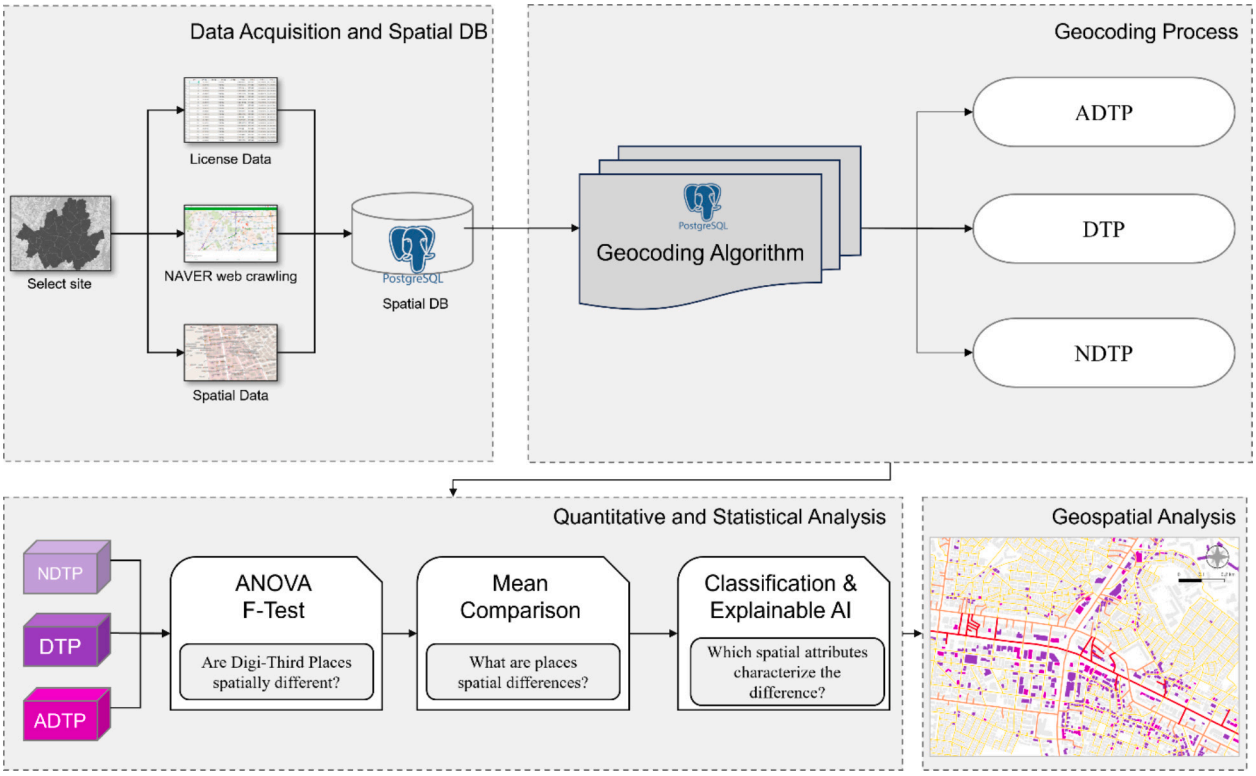


Fig. 1. Research process.

represents the most finely resolved spatial unit within the Korean address system, while "place type" corresponds to the specific places in an activity category, such as "restaurant" or "beauty shop." Subsequently, from the pool of DTP data, we extracted ADTP data by querying places that provide a web link for reservation services. Notably, we excluded places that offer delivery services to clarify the definition of Third Places in this study: physical spaces that provide opportunities to visit and participate in social activities.

The NDTP dataset, which was unsearchable in NAVER Place, was retrieved from licensing data provided by the Ministry of the Interior and Safety (MOIS) (Ministry of the Interior and Safety, 2023a). Eating places constituted the largest proportion, encompassing a wide spectrum of food businesses such as restaurants, cafes, bakeries, and pubs. Clubs, which cater to individuals seeking to participate in hobbies and group activities, encompassed physical activities or social clubs. This category included sports clubs dedicated to regular lessons in martial arts and fitness clubs. Within the realm of Retail, we focused on local markets and neighborhood facilities for local residents. We gathered data for beauty shops and barber shops, as online reservations for such facilities have increasingly proliferated. Education included private tutoring facilities providing language or art instruction, reflecting the prevalent trend in South Korea of individuals seeking amusement through learning. Unlike the other categories, data for Education were gathered from educational facility data provided by the Seoul Metropolitan Office of Education (Seoul Metropolitan Office of Education, 2023). To clarify the definition of Third Places in this study, we excluded private tutoring facilities that exclusively targeted students preparing for entrance exams.

To represent the spatial characteristics of Third Places, we collected spatial attributes referred to in relevant studies on the location decisions of retail shops or restaurants (Erbıyık et al., 2012; Huff, 1966; Lin et al., 2018; Roig-Tierno et al., 2013; Tinessa et al., 2020; Tzeng et al., 2002). To systemically elucidate the spatial characteristics of Third Places, we delineated three essential dimensions: 1) location, 2) building, and 3) zoning plan. These dimensions served as the basis for our comprehensive analysis of the spatial differences evident within the NDTP, DTP, and ADTP categories. Each dimension has representative attributes that are documented in Table 1. The spatial attributes and licensing data were integrated using PostgreSQL.

We set the cutoff date, the date when we ended the crawl of all Third Place information in NAVER Place, at July 14, 2023. We calculated the building age by counting the number of days from the built completion date to the cutoff date.

The disparate data sources for DTP, ADTP, and NDTP inevitably engendered inconsistencies in the spatial data, necessitating data preprocessing to synchronize the geographical units.

3.3. Geocoding process

The geocoding process, illustrated in Fig. 4, ensured the consistency of the NDTP and DTP data, which originated from different sources, by aligning them with identical building geometries. In instances where the address attribute was unusable, the geocoding process employed the coordinate attribute to join the place data to the nearest building geometry.

When a building possessed only licensing data without any of the places crawled from NAVER, the geocoding process classified the building as NDTP. However, when both NAVER Place data and licensing data coexisted for an identical building, the geocoding algorithm discerned whether the licensing data were searchable in NAVER. To establish this differentiation, the geocoding process employed a text distance function provided by PostgreSQL. The function measured the dissimilarity between the business name associated with the licensing data and the names of NAVER places located in an identical building. If the calculated text distance fell below 0.2, indicating a significant textual dissimilarity between the two names, the licensing data in the building were considered unsearchable in NAVER Place; therefore, the place was classified as NDTP. Conversely, if the place was searchable in the NAVER dataset, it fell into the DTP or ADTP groups. The criteria for the text distance were established through empirical observations made

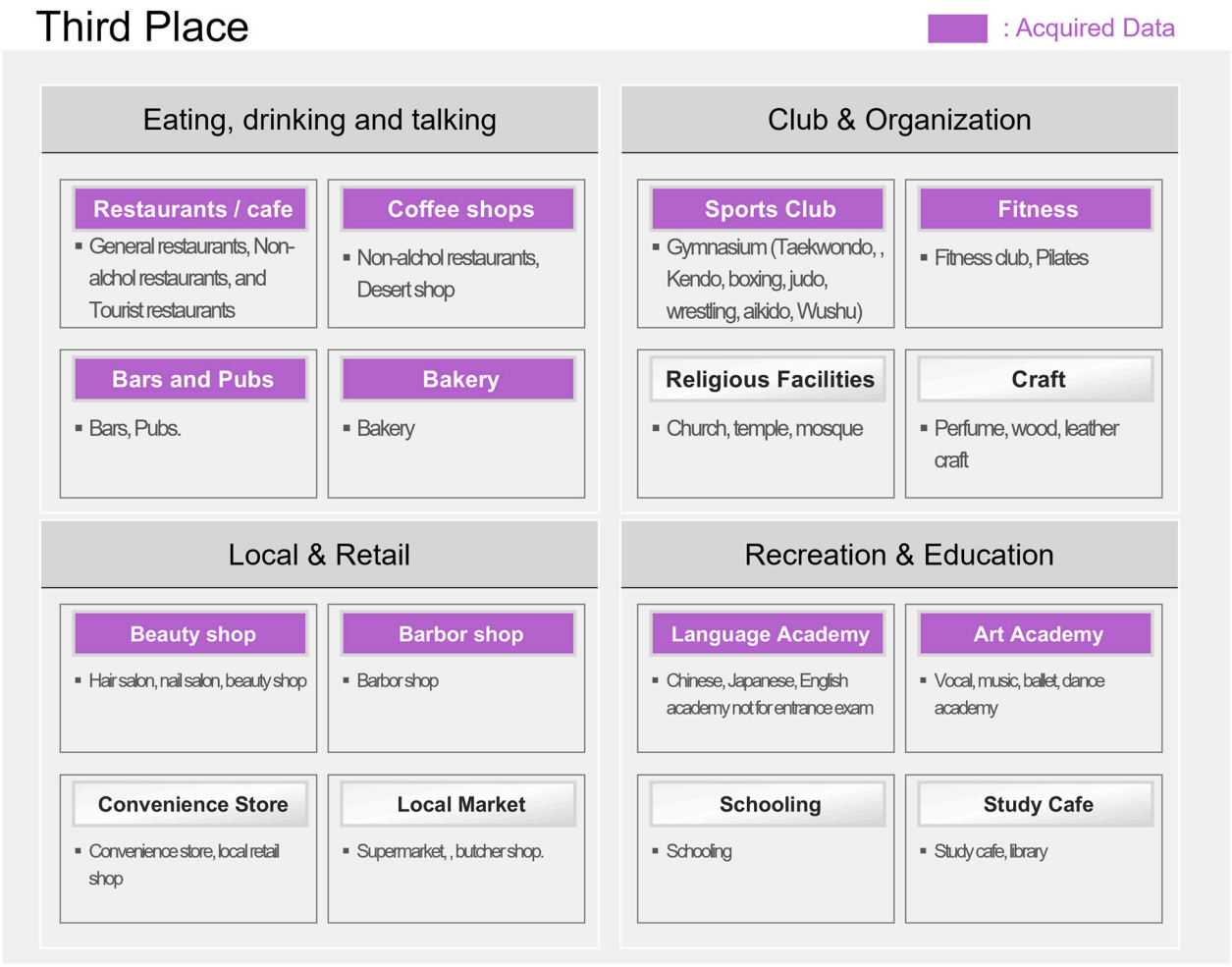


Fig. 2. Categories of Third Places by types of activities.

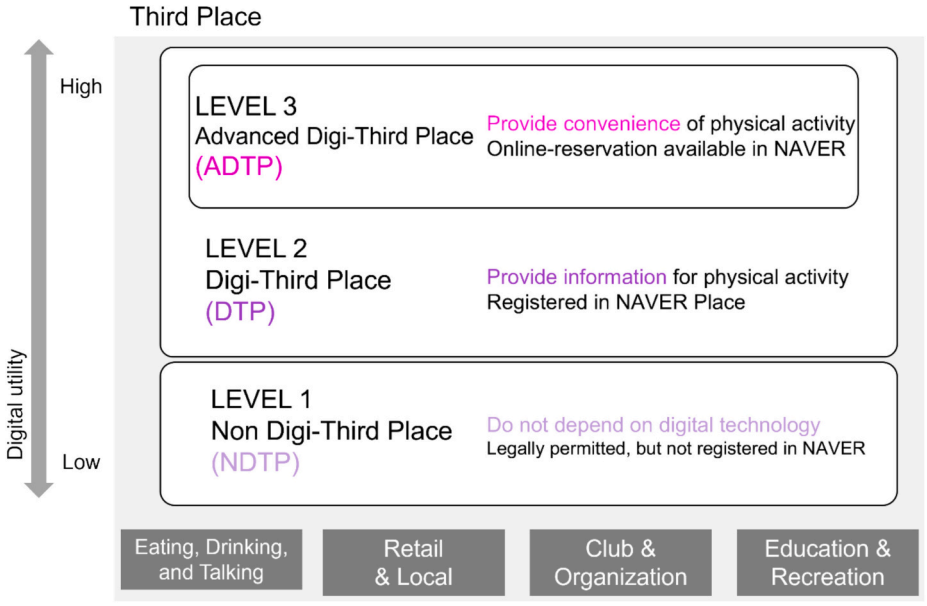


Fig. 3. Classification of Third Places by the level of digital utility.

Table 1
Spatial attributes of Third Places.

Dimension	Attributes	Description	Mean	Std	(Min, Max)	Source
Location	Hierarchy of the nearest road (Road Hierarchy)	0: Main trunk way, 1: Subsidiary trunk way, 2: collector road, 3: local road	3.727	0.484	(1, 4)	Road and Building Information (Ministry of the Interior and Safety, 2023b)
	Dependency class of the nearest road (Road Dependency)	0: Main Street, 1: 1st minor street, 2: 2nd minor street	0.357	0.505	(0, 2)	
	Width of the nearest road (Road Width)	Width of the road (m)	9.379	8.286	(1.0, 110)	
	Distance to the nearest road (Dist. Road)	Euclidean distance to the road geometry (m)	9.530	20.286	(0.0, 893.055)	
	Distance to the nearest apartment building (Dist. APT)	Euclidean distance to the nearest apartment building (m)	108.923	127.881	(0.0, 2275.07)	
	Distance to the nearest central business district (Dist. CBD)	Euclidean distance to the nearest stations in the CBD (m)	5558.183	3325.498	(0.0, 15,766.255)	
	Distance to the nearest subway station (Dist. Station)	Euclidean distance to the nearest subway station (m)	449.183	350.140	(0.0, 3455.817)	
Building	Number of floors (Num. Floors)	Number of floors of building	5.991	6.948	(0, 123)	Building information (Ministry of the Interior and Safety, 2023b)
	Number of underground floors (Num. Underground Floors)	Number of basement floors of building	1.343	1.593	(0, 23)	
	Total floor area (Total Floor Area)	Total floor area of building	1105.373	5074.380	(2.175, 85,325.840)	
	In residential building use	1: In residential building, 0: else	0.277	0.447	(0, 1)	
	In neighborhood building use	1: In neighborhood building, 0: else	0.531	0.499	(0, 1)	
	In commercial building use	1: In commercial building, 0: else	0.054	0.226	(0, 1)	
	In office building use	1. In office building, 0: else	0.079	0.269	(0, 1)	
	In industrial building use	In industrial building, 0: else	0.014	0.119	(0, 1)	
	Days after constructed (Building Age)	Days after built until data acquired (days)	10,794.254	5546.915	(497, 56,297)	Building information (Ministry of the Interior and Safety, 2023b; Ministry of the Land Infrastructure and Transport, 2023a)
Zoning Plan	In exclusive residential zone	1: In exclusive residential zone, 0: else	0.003	0.053	(0, 1)	Building information (Ministry of the Interior and Safety, 2023b),
	In general residential zone	1: In general residential zone, 0: else	0.002	0.040	(0, 1)	Zoning plan (Ministry of the Land Infrastructure and Transport, 2023b)
	In semiresidential zone	1: In semiresidential zone, 0: else	0.086	0.281	(0, 1)	
	In commercial zone	1: In commercial zone, 0: else	0.200	0.400	(0, 1)	
	In industrial zone	1: In industrial zone, 0: else	0.051	0.219	(0, 1)	
	In green zone	1: In green zone, 0: else	0.007	0.085	(0, 1)	

by the authors.

3.4. Method

3.4.1. ANOVA F-test

To address the first research subquestion, we employed the analysis of variance (ANOVA) F-test. The ANOVA F-test is a statistical tool that assesses whether the differences among the means of multiple groups are statistically significant. Pertinent studies have conducted ANOVA F-tests to determine the statistical significance of mean differences among multiple groups of social settings or survey results (Koenig et al., 1996; Lee, 2022). When the calculated p-value falls below a designated significance level, it suggests that the mean values of the groups exhibit statistically significant variation. In this study, we employed the ANOVA F-test to determine whether NDTP, DTP, and ADTP exhibited distinct spatial characteristics. The implementation of the statistical analysis was executed using the "statsmodel" module in Python.

3.4.2. Mean comparison

To investigate the second research subquestion, elucidating the spatial difference among Third Places in terms of digital utility, we employed a straightforward comparative analysis. This analysis involved the examination of the mean and standard deviation for each of the distinct groups. The mean comparison unraveled valuable insights into the unique spatial characteristics associated with each group in comparison to the others.

3.4.3. Classification and explainable AI

We employed various multiclass classifier models encompassing both linear and nonlinear models to address the final research subquestion. Initially, logistic regression, which is a linear model, was used to classify DTP locations based on their spatial attributes. The outcomes of the logistic regression provided valuable insights into the spatial characteristics that exerted a substantial influence on the differentiation among the NDTP, DTP, and ADTP groups.

In our pursuit of a more robust model, we further delved into the performance assessment of several classifiers, including nonlinear models. Comparing the performance of the classifiers, including the naïve Bayesian (NB) classifier, support vector machine (SVM), and random forest (RF) classifier, we evaluated the models' confusion matrices and accuracy scores.

Notably, machine learning-based approaches, including RF, have

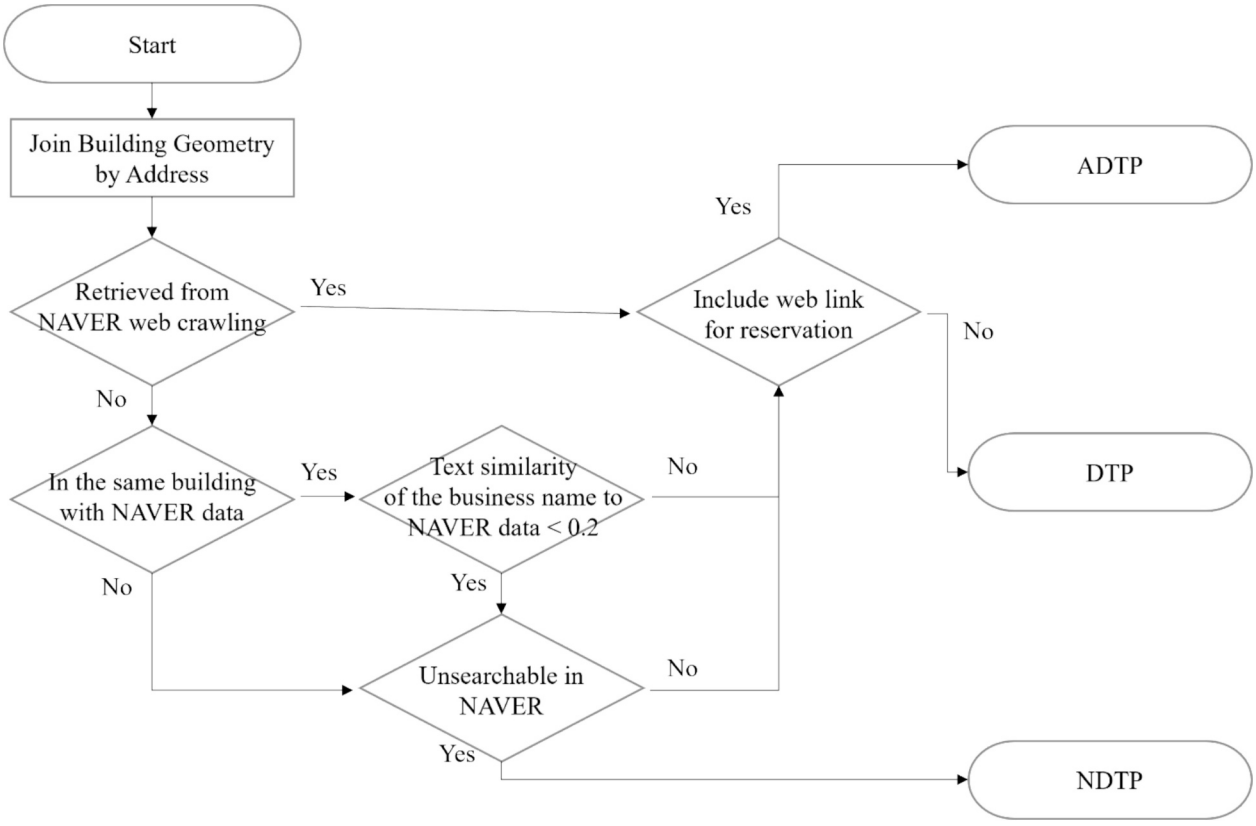


Fig. 4. Diagram of the geocoding process.

faced criticism due to the inherent challenge of explainability within the model. To overcome this concern and enhance the interpretability of our findings, we adopted an explainable AI approach, specifically shapely additive explanations (SHAP), a method to explain individual predictions based on game theory (Lundberg & Lee, 2017; Strumbelj & Kononenko, 2014). SHAP values quantify the explanatory influence of each variable within a regression model by systematically approximating the simple model with interpretable binary inputs, adhering to the principles of game theory (Lundberg & Lee, 2016). The approximation calculates SHAP values based on the average expected marginal contribution of each variable. SHAP has been extensively applied to represent the explanatory power of individual variables within analysis models rooted in machine learning, as prior studies have shown (Kim & Kim, 2022; Simini et al., 2021). Leveraging explainable AI through SHAP offers a pathway to understand the contributions of each spatial attribute to the DTP characteristics based on AI that is transparent, interpretable, and trustworthy.

3.5. Geospatial analysis

To understand the results of the quantitative model in the urban context, we conducted a geospatial analysis, focusing on the principal spatial attributes. This in-depth exploration aimed to elucidate the intricate relationships among the key spatial factors identified through the classifier model analysis and the empirical phenomena stemming from NDTP, DTP, and ADTP within the urban environment. To achieve this, we interpreted the result of geospatial analysis, with a specific focus on notable sites in Seoul. Through geospatial analysis, this study scrutinized the physical manifestations of our findings in urban environments.

4. Results

4.1. Basic statistics

Table 2 provides a comprehensive overview of the distribution of places across different activity categories and their levels of digital utility. The results show the extent to which people rely on digital technology to participate in social activities.

According to Table 2, DTP is most prevalent in the Eating category. The proportion of NDTP is lowest in the Retail category. The Retail and Club categories exhibit a relatively high proportion of ADTP, indicating that customers frequently utilize reservation services before visiting such places. Conversely, the Education category shows the lowest proportion of ADTP and the highest proportion of DTP.

The basic statistics revealed that 1) 76 % of Third Places in Seoul appeared to be reliant on digital technology for their businesses, 2) Retail and Club places appeared to be more proactive in leveraging reservation services, and 3) Education places offered their information through an online platform rather than providing reservation services.

Table 2
Number and proportion of places depending on digital utility and type of activity.

	Eating	Retail	Club	Education	Total
NDTP	38,729 (24.68 %)	5895 (17.95 %)	1655 (27.98 %)	1445 (37.18 %)	47,724
DTP	107,973 (68.79 %)	20,914 (63.68 %)	3235 (54.70 %)	2324 (59.79 %)	134,446
ADTP	10,253 (6.53 %)	6034 (18.37 %)	1024 (17.31 %)	118 (3.03 %)	17,429
Total	156,955 (100 %)	32,843 (100 %)	5914 (100 %)	3887 (100 %)	199,599

4.2. Statistical significance of differences in spatial attributes

To determine whether the level of digital utility significantly differed from the spatial variables, this study employed the ANOVA F-test. Table 3 tabulates the results of the ANOVA F-test for the entire Third Places dataset without categorizing social activities. In a one-way ANOVA F-test, a p-value < 0.05 is the significance level for rejecting the null hypothesis that all the attributes have no statistical significance in the mean difference. As presented in Table 3, except for a Dist. Station attribute in the location dimension, and two attributes in the zoning dimension such as exclusive residential, general residential zones, most attributes exhibited significant differences, implying that the majority of spatial characteristics vary significantly based on the level of digital utility.

Table 4, which presents the results of the ANOVA F-tests conducted for each activity category, shows that the majority of spatial attributes exhibited significant differences regardless of the activity categories. Third places in the Eating category have significantly different spatial characteristics for entire variables, except for whether they were located within an exclusive residential zoning. Retail places also showed no significant variation in terms of Road Dependency, Num. Floors, or whether they are located in residential or commercial buildings. Among the zoning plan variables, only exclusive residential and semi-residential zones were statistically significant. Club places exhibited statistical significance for several location variables, such as Road Hierarchy, Dist. Road, and Dist. CBD, and building sizes, including Total Floor Area or Num. Floor. Last, the difference in Road. Dependency, Dist. CBD, Dist. Station and Total Floor Area, and building uses including neighborhood or industrial building variables, was statistically significant in the analysis of education places.

In summary, Tables 3 and 4 show that spatial differences vary by digital utility depending on the type of activity. The spatial variables of the location dimension have statistical significance, especially Road Hierarchy, Dist. Road, Dist. CBD. In terms of building dimension, Total Floor Area, Num. Floors, Num. Underground Floors, and building age exhibited statistical significance. Among building use variables, neighborhood, office, and industrial building use have statistical significance in most activity categories. However, the zoning plan did not exhibit significant differences in spatial variables according to the classification of Third Places.

Table 3
ANOVA F-test results depending on the level of digital utility without activity category.

Dimension	Attributes	F	p-value
Location	Road Hierarchy	1153.220	<0.05 *
	Road Dependency	46.90	<0.05 *
	Road Width	792.247	<0.05 *
	Dist. Road	1032.961	<0.05 *
	Dist. APT	126.767	<0.05 *
	Dist. CBD	30.373	<0.05 *
	Dist. Station	3.460	0.06
Building	Num. Floor	89.421	<0.05 *
	Num. Underground Floor	344.892	<0.05 *
	Total Floor Area	879.500	<0.05 *
	Residential building use	321.135	<0.05 *
	Neighborhood building use	196.346	<0.05 *
	Commercial building use	941.684	<0.05 *
	Office building use	115.407	<0.05 *
	Industrial building use	58.132	<0.05 *
	Building Age	32.975	<0.05 *
Zoning Plan	Exclusive residential zone	1.233	0.267
	General residential zone	0.171	0.679
	Semiresidential zone	19.597	<0.05 *
	Commercial zone	864.154	<0.05 *
	Industrial zone	36.402	<0.05 *
	Green zone	241.196	<0.05 *

Table 4
ANOVA F-test results depending on the level of digital utility and activity category.

Dimension	Attributes	Eating F (p-value)	Retail F (p-value)	Club F (p-value)	Education F (p-value)
Location	Road Hierarchy	948.308 (<0.05 *)	790.683 (<0.05 *)	8.453 (<0.05 *)	0.699 (0.403)
	Road Dependency	56.561 (<0.05 *)	0.193 (0.661)	0.009 (0.924)	6.799 (<0.05 *)
	Road Width	723.086 (<0.05 *)	73.286 (<0.05 *)	1.322 (0.250)	1.509 (0.219)
	Dist. Road	838.601 (<0.05 *)	212.116 (<0.05 *)	74.825 (<0.05 *)	0.105 (0.746)
	Dist. APT	49.746 (<0.05 *)	7.509 (<0.05 *)	0.860 (0.357)	0.054 (0.817)
	Dist. CBD	20.925 (<0.05 *)	96.478 (<0.05 *)	4.900 (<0.05 *)	11.316 (<0.05 *)
	Dist. Station	10.605 (<0.05 *)	37.187 (<0.05 *)	0.427 (0.513)	4.212 (<0.05 *)
Building	Num. Floor	143.500 (<0.05 *)	0.096 (0.752)	4.368 (<0.05 *)	3.992 (<0.05 *)
	Num. Underground Floor	327.993 (<0.05 *)	10.350 (<0.05 *)	7.231 (<0.05 *)	19.683 (<0.05 *)
	Total Floor Area	805.343 (<0.05 *)	11.525 (<0.05 *)	0.040 (0.842)	6.343 (<0.05 *)
	Residential building use	200.116 (<0.05 *)	2.505 (0.113)	0.028 (0.867)	1.350 (0.245)
	Neighborhood building use	255.149 (<0.05 *)	41.989 (<0.05 *)	0.930 (0.335)	7.988 (<0.05 *)
	Commercial building use	893.329 (<0.05 *)	25.155 (0.052)	1.801 (0.179)	0.001 (0.972)
	Office building use	70.401 (<0.05 *)	27.690 (<0.05 *)	0.008 (0.931)	8.930 (<0.05 *)
	Industrial building use	82.864 (<0.05 *)	13.590 (<0.05 *)	0.286 (0.593)	6.054 (<0.05 *)
	Building Age	14.887 (<0.05 *)	244.427 (<0.05 *)	34.218 (<0.05 *)	1.706 (0.192)
Zoning Plan	Exclusive residential zone	4.197 (<0.05 *)	7.515 (<0.05 *)	0.098 (0.754)	0.176 (0.675)
	General residential zone	0.077 (0.781)	1.092 (0.296)	0.272 (0.602)	0.408 (0.523)
	Semi-residential zone	6.726 (<0.05 *)	17.657 (<0.05 *)	4.333 (<0.05 *)	1.016 (0.314)
	Commercial zone	840.137 (<0.05 *)	3.797 (0.051)	15.772 (<0.05 *)	5.465 (<0.05 *)
	Industrial zone	51.044 (<0.05 *)	3.259 (0.071)	1.224 (0.286)	0.327 (0.567)
	Green zone	220.643 (<0.05 *)	0.250 (0.617)	0.107 (0.744)	2.227 (0.136)

4.3. NDTP, DTP, and ADTP spatial characteristics

In this section, we scrutinize the spatial characteristics of Third Places according to the level of digital utility and social activity, comparing the mean and standard deviation of spatial attributes. Table 5

Table 5
Mean comparison of spatial attributes (bold character: p-value < 0.05).

Attributes	Eating Mean (Std)			Retail Mean (Std)			Club Mean (Std)			Education Mean (Std)		
Digital Utility	NDTP	DTP	ADTP	NDTP	DTP	ADTP	NDTP	DTP	ADTP	NDTP	DTP	ADTP
Road Hierarchy	**3.673**	**3.733**	**3.833**	**3.695**	**3.750**	**3.811**	**3.657**	**3.668**	**3.721**	3.703	3.675	3.754
	(0.525)	**(0.478)**	**(0.400)**	**(0.497)**	**(0.462)**	**(0.418)**	**(0.520)**	**(0.508)**	**(0.489)**	(0.521)	(0.518)	(0.505)
Road Dependency	**0.342**	**0.370**	**0.363**	0.345	0.356	0.350	0.296	0.300	0.293	**0.265**	**0.294**	**0.373**
	(0.500)	**(0.510)**	**(0.509)**	(0.500)	(0.500)	(0.497)	(0.472)	(0.479)	(0.474)	**(0.451)**	**(0.463)**	**(0.503)**
Road Width	**10.300**	**9.135**	**8.324**	**9.795**	**8.946**	**8.577**	10.805	10.764	10.357	10.815	10.750	8.700
	(9.568)	**(7.963)**	**(6.612)**	**(8.544)**	**(7.734)**	**(6.960)**	(9.054)	(8.989)	(8.267)	(9.478)	(9.501)	(7.781)
Dist. Road	**12.391**	**8.699**	**7.707**	**12.231**	**8.645**	**8.043**	**13.570**	**10.751**	**9.970**	8.944	9.210	7.089
	(27.746)	**(19.080)**	**(13.960)**	**(22.517**	**(14.296)**	**(11.013)**	**(29.510)**	**(13.703)**	**(13.384)**	(7.374)	(9.253)	(5.301)
Dist. APT	**122.520**	**114.689**	**118.730**	**80.922**	**78.589**	**85.345**	75.549	65.807	74.824	89.814	87.385	107.308
	(142.374)	**(132.351)**	**(133.783)**	**(97.614)**	**(86.994)**	**(89.544)**	(95.277)	(82.047)	(94.567)	(101.968)	(112.750)	(118.806)
Dist. CBD	**5196.374**	**5552.326**	**4882.489**	**6018.174**	**6509.047**	**5440.871**	**5901.371**	**6402.559**	**5411.695**	**4341.710**	**4903.050**	**3881.693**
	(3314.982)	**(3315.404)**	**(3154.732)**	**(3284.074)**	**(3258.088)**	**(3201.350)**	**(3412.147)**	**(3357.549)**	**(3260.813)**	**(3250.512)**	**(3368.659)**	**(3041.722)**
Dist. Station	**431.313**	**448.489**	**423.153**	**451.032**	**499.760**	**412.436**	477.562	514.812	455.045	**438.134**	**432.886**	**331.409**
	(341.423)	**(353.389)**	**(322.121)**	**(335.272)**	**(367.797)**	**(316.517)**	(357.264)	(390.215)	(373.692)	**(297.118)**	**(319.202)**	**(170.710)**
Num. Floor	**6.428**	**5.699**	**6.159**	6.932	5.785	6.876	**7.036**	**6.327**	**7.840**	**6.637**	**6.311**	**6.186**
	(7.604)	**(6.738)**	**(7.239)**	(7.693)	(6.688)	(7.259)	**(6.520)**	**(5.931)**	**(6.601)**	**(4.560)**	**(5.253)**	**(3.088)**
Num. Underground Floor	**1.513**	**1.288**	**1.372**	**1.423**	**1.179**	**1.331**	1.644	1.362	1.919	**1.711**	**1.523**	**1.347**
	(1.756)	**(1.553)**	**(1.657)**	**(1.673)**	**(1.421)**	**(1.595)**	(1.684)	(1.369)	(1.838)	**(1.474)**	**(1.369**	**(1.007)**
Total Floor Area	**1961.952**	**1024.386**	**732.226**	**752.123**	**580.773**	**614.212**	755.517	605.056	817.654	**623.055**	**557.044**	**403.009**
	(8025.253)	**(4657.974)**	**(2820.785)**	**(2194.225)**	**(2190.603)**	**(2202.326)**	(2288.456)	(1589.749)	(1602.791)	**(1264.458)**	**(894.230)**	**(495.537)**
Residential building	**0.233**	**0.268**	**0.283**	0.367	0.409	0.353	0.176	0.201	0.166	0.101	0.102	0.025
	(0.423)	**(0.443)**	**(0.450)**	(0.482)	(0.492)	(0.478)	(0.381)	(0.401)	(0.372)	(0.301)	(0.302)	(0.158)
Neighborhood building	**0.495**	**0.548**	**0.547**	**0.460**	**0.474**	**0.519**	0.599	0.633	0.609	**0.655**	**0.679**	**0.814**
	(0.500)	**(0.498)**	**(0.498)**	**(0.498)**	**(0.499)**	**(0.500)**	(0.490)	(0.482)	(0.488)	**(0.476)**	**(0.467)**	**(0.391)**
Commercial building	**0.092**	**0.052**	**0.033**	0.043	0.030	0.027	0.029	0.022	0.041	0.028	0.027	0.034
	(0.288)	**(0.222)**	**(0.178)**	(0.202)	(0.169)	(0.161)	(0.168)	(0.147)	(0.198)	(0.164)	(0.161)	(0.182)
Office building	**0.091**	**0.074**	**0.081**	**0.097**	**0.062**	**0.072**	0.115	0.094	0.123	**0.131**	**0.099**	**0.093**
	(0.288)	**(0.262)**	**(0.272)**	**(0.296)**	**(0.242)**	**(0.258)**	(0.320)	(0.292)	(0.329)	**(0.337)**	**(0.299)**	**(0.292)**
Industrial building	**0.011**	**0.018**	**0.021**	**0.006**	**0.006**	**0.012**	0.013	0.008	0.017	**0.018**	**0.009**	**0.092**
	(0.105)	**(0.132)**	**(0.142)**	**(0.020)**	**(0.077)**	**(0.109)**	(0.112)	(0.089)	(0.128)	**(0.133)**	**(0.092)**	**(0.122)**
Building Age	**10,680.298**	**11,067.450**	**10,431.315**	**10,502.024**	**10,764.478**	**9016.035**	**10,177.368**	**10,091.286**	**9024.983**	9810.858	10,016.651	10,004.492
	(5715.653)	**(5637.885)**	**(5558.127)**	**(5333.428)**	**(5147.351)**	**(5355.486)**	**(4482.923)**	**(4473.512)**	**(4390.382)**	(4479.036)	(4453.347)	(4229.371)
Exclusive residential zone	**0.004**	**0.003**	**0.003**	**0.001**	**0.002**	**0.003**	0.002	0.002	0.002	0.001	0.000	0.000
	(0.060)	**(0.054)**	**(0.052)**	**(0.034)**	**(0.044)**	**(0.059)**	(0.049)	(0.043)	(0.044)	(0.026)	(0.021)	(0.000)
General residential zone	**0.001**	**0.002**	**0.001**	**0.002**	**0.003**	**0.003**	**0.002**	**0.002**	**0.001**	0.000	0.000	0.000
	(0.038)	**(0.040)**	**(0.030)**	**(0.043)**	**(0.051)**	**(0.032)**	**(0.043)**	**(0.039)**	**(0.031)**	(0.000)	(0.021)	(0.000)
Semiresidential zone	**0.080**	**0.090**	**0.077**	**0.085**	**0.074**	**0.106**	**0.099**	**0.110**	**0.125**	0.075	0.090	0.059
	(0.271)	**(0.286)**	**(0.267)**	**(0.279)**	**(0.261)**	**(0.308)**	**(0.299)**	**(0.313)**	**(0.331)**	(0.263)	(0.286)	(0.237)
Commercial zone	**0.273**	**0.198**	**0.188**	0.169	0.120	0.156	**0.147**	**0.116**	**0.219**	**0.179**	**0.196**	**0.288**
	(0.445)	**(0.399)**	**(0.390)**	(0.375)	(0.325)	(0.363)	**(0.354)**	**(0.320)**	**(0.331)**	**(0.384)**	**(0.397)**	**(0.455)**
Industrial zone	**0.045**	**0.055**	**0.057**	0.042	0.037	0.048	0.051	0.052	0.062	0.042	0.039	0.034
	(0.208)	**(0.228)**	**(0.232)**	(0.200)	(0.188)	(0.214)	(0.220)	(0.223)	(0.240)	(0.201)	(0.194)	(0.182)
Green zone	**0.015**	**0.007**	**0.004**	0.003	0.001	0.002	0.003	0.003	0.004	0.003	0.001	0.000
	(0.121)	**(0.083)**	**(0.062)**	(0.054)	(0.033)	(0.050)	(0.055)	(0.053)	(0.062)	(0.053)	(0.029)	(0.000)

presents the mean and standard deviation for spatial attributes, highlighting statistically significant differences according to the ANOVA F-test in Table 4.

Notably, Road Hierarchy displays consistent trends across Eating, Retail, and Club activity categories. Higher levels of digital utility correspond to a higher level of road hierarchy, indicating that places that leverage digital utility tend to be located on local or collector roads rather than on main and subsidiary trunk ways. The Road Width observation that places with high digital utility are located closer to narrower roads supports the results of Road Hierarchy.

Across all activity categories, Third Places exhibit notably distinct spatial characteristics in location dimensions. For instance, places with lower levels of digital utility are farther from the nearest road, except Third Places in Education category. Specifically, most buildings farther from the road are located within collective buildings, such as apartment complexes or university campuses, for which road geometries are not represented in open-source data. In terms of Dist. CBD, there are notable differences among all activity categories. Specifically, ADTP shows significant proximity to the CBD. Interestingly, DTP locations in all categories are generally farther from the CBD. In terms of Dist. APT, Eating and Retail categories show that DTP locations tend to be closer to apartment buildings; however, the tendency is not remarkable compared to the standard deviation.

In the building dimensions, most of the attributes, such as Num. Floor, Total Floor Area, etc., were statistically significant in the variables related to building size. In particular, it was consistently found that the Total Floor Area of the building in which the NDTP is located is the largest. It was also found that the Building Age in which the ADTP is located is the lowest among the remaining activity categories, with the exception of Education. The variables corresponding to building use and zoning plan are also statistically significant, but considering the large standard deviations, the variables related to building or land use seem not noteworthy.

4.4. Understanding the spatial differences using classification and explainable AI

4.4.1. Multinomial logit regression

To assess the impact of each spatial attribute on differences based on digital utility, this study first employed multinomial Logit regression (MNL). Prior to conducting the MNL, we standardized several attributes, including distance-related attributes, Total Floor Area, and Building Age. We removed the neighborhood building attribute because of multicollinearity.

We conducted MNL for all Third Places activity categories by the level of digital utility, as shown in Table 6. We separated the dataset into a training set and a dataset at a ratio of 8:2, with samples randomly selected for analysis. The odds column, which is an exponential of the coefficient column, indicates that the increment of a unit of an independent variable increases the probability of changing the classification result. The model's performance is assessed using a pseudo-R square value of 0.018 and an accuracy rate of 67.198 %.

Table 7 tabulates the performance of the MNL we developed. The MNL model used in this study exhibits certain weaknesses, primarily due to its low pseudo R-square value. Moreover, the model struggles with classifying ADTP accurately, indicating that MNL is not suitable for capturing the spatial characteristics of ADTP effectively. To clearly distinguish spatial characteristics based on digital utility, the classifier should be chosen based on performance comparison.

4.4.2. Comparison of classifiers

To identify the most suitable classifier model for elucidating the spatial characteristics distinguished by digital utility, we compared the performances of four classifiers: MNL, NB, SVM, and RF. Table 8 presents the performance of these classifiers. Among the classifiers, the RF model exhibits the best performance in terms of accuracy, precision, and

Table 7
Confusion matrix and performance of the multinomial logit regression classifier.

Predicted label	True label		
	NDTP	DTP	ADTP
NDTP	92	87	1
DTP	9488	26,741	3522
ADTP	0	0	0
Precision	51.111	67.271	0.0
Recall	0.960	99.676	0.0
Accuracy	67.198 %		

Table 6
Multinomial logit regression for the level of digital utility of Third Places.

	NDTP to DTP			NDTP to ADTP		
Digital utility	Coefficient	Odds	p-value	Coefficients	Odds	p-value
Constant	0.473	1.604	<0.005 ***	−2.975	0.051	<0.005 ***
Road Hierarchy	0.145	1.156	<0.005 ***	0.588	1.801	<0.005 ***
Road Dependency	0.052	1.053	<0.005 ***	0.096	1.101	<0.005 ***
Road Width	0.003	1.003	<0.005 ***	0.009	1.009	<0.005 ***
Dist. Road	−0.132	0.876	<0.005 ***	−0.231	0.794	<0.005 ***
Dist. APT	0.018	1.019	<0.005 ***	−0.024	0.977	0.018
Dist. CBD	0.098	1.103	<0.005 ***	−0.091	0.913	<0.005 ***
Dist. Station	0.027	1.027	<0.005 ***	−0.080	0.923	<0.005 ***
Num. Floor	0.001	1.001	0.553	0.011	1.012	0.005 **
Num. Underground Floor	−0.031	0.969	<0.005 ***	−0.022	0.979	<0.005 ***
Total Floor Area	−0.052	0.950	0.712	−0.207	0.813	<0.005 ***
Residential building	0.124	1.132	<0.005 ***	0.113	1.120	<0.005 ***
Commercial building	−0.345	0.708	<0.005 ***	−0.672	0.511	<0.005 ***
Office building	−0.072	0.931	<0.005 ***	−0.135	0.874	<0.005 ***
Industrial building	0.394	1.483	<0.005 ***	0.454	1.575	<0.005 ***
Building Age	0.000	1.000	<0.005 ***	0.000	1.000	<0.005 ***
Exclusive residential zone	−0.318	0.728	<0.005 ***	−0.206	0.814	0.208
General residential zone	−0.123	0.884	0.371	−0.544	0.580	0.005 **
Semiresidential zone	0.020	1.020	0.315	0.026	1.026	0.430
Commercial zone	−0.175	0.840	<0.005 ***	−0.356	0.701	<0.005 ***
Industrial zone	0.138	1.149	<0.005 ***	0.091	1.095	<0.005 ***
Green zone	−0.452	0.637	<0.005 ***	−0.591	0.554	<0.005 ***
Pseudo R-square	0.018					
Accuracy	67.198 %					

Table 8
Comparison of model performance.

Models	MNL			NB			SVM			RF		
Predicted label	True label											
	NDTP	DTP	ADTP	NDTP	DTP	ADTP	NDTP	DTP	ADTP	NDTP	DTP	ADTP
NDTP	96	91	1	1613	2654	315	0	0	0	1627	1502	149
DTP	9484	26,737	3522	7674	23,445	3071	9580	26,828	3523	7926	25,203	3327
ADTP	0	0	0	293	729	137	0	0	0	27	123	47
Precision	51.064	67.275	0.0	35.203	68.573	11.821	0.0	67.186	0.0	49.634	69.133	23.858
Recall	1.002	99.661	0.0	16.837	87.39	3.889	0.0	100	0.0	16.983	93.943	1.334
Accuracy	67.198 %			63.096			67.186			67.309		

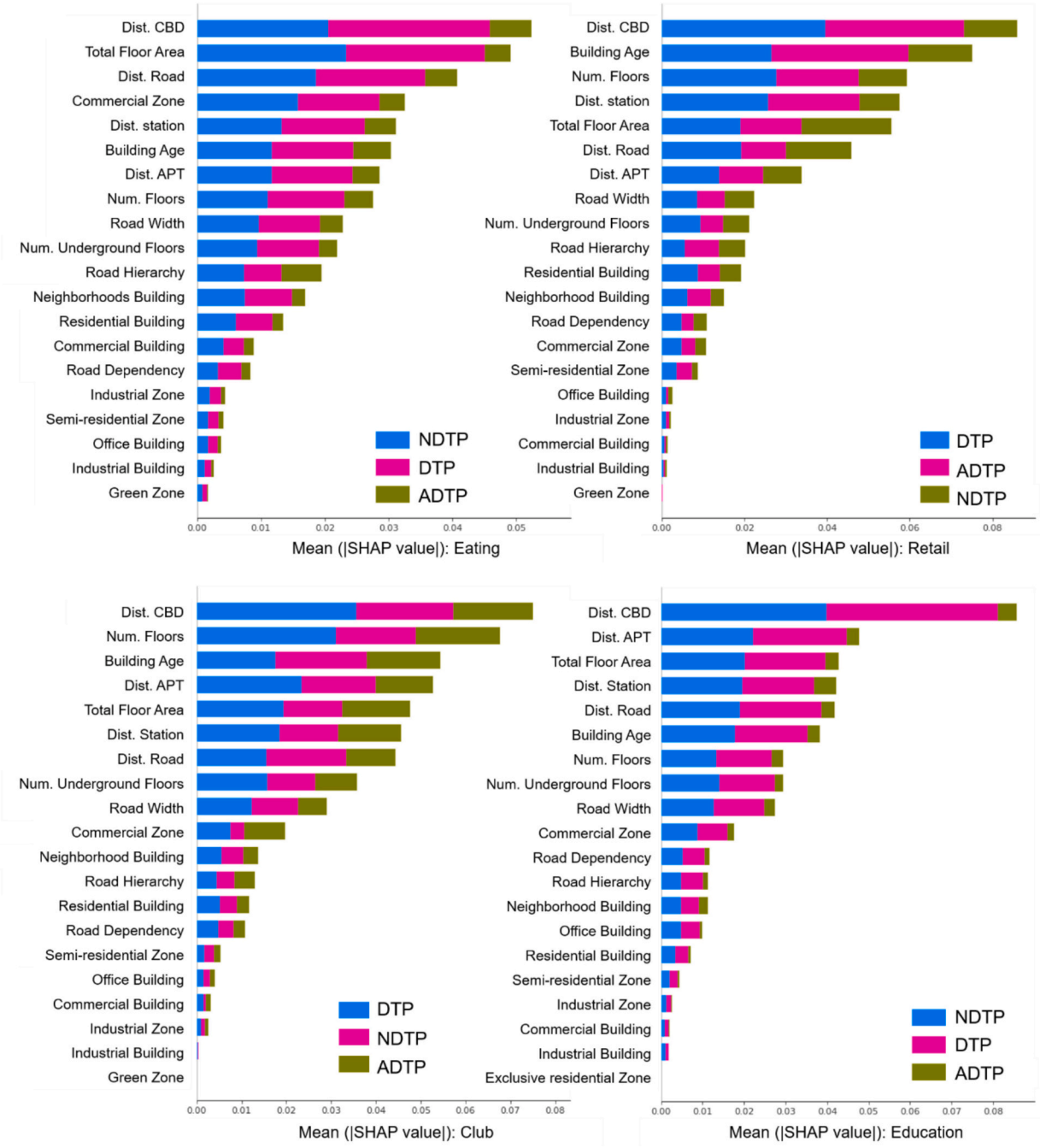


Fig. 5. Feature importance by SHAP value for RF classification.

recall values.

4.4.3. SHAP analysis

To understand the nonlinear influence of spatial attributes in RF, we conducted SHAP analysis for each activity category. Fig. 5 demonstrates the average impact on the classifier model output magnitude for each activity category. Across all categories, the remarkable attributes Dist. CBD, Building Size (Total Floor Area, Num. Floors, and Num. Underground Floors), Dist. Station, Building Age, and Dist. Road. exhibit critical influence on the classification according to the level of digital utility.

Figs. 6 and 7 depict the influence of spatial attributes, represented by color, on the classification of each category. The y-axis represents the attributes that have a significant impact on the classification into each class, with attributes higher on the axis having a greater influence. The color placement along the x-axis signifies a positive or negative impact, with factors to the right indicating a positive influence. For instance, a red dot positioned to the right along the x-axis signifies that as the value of that feature increases, there is a positive correlation with the probability of classifying it into the particular class. Importantly, the interpretation should focus on feature importance rather than statistical significance.

In the analysis of NDTP for Eating places in Fig. 6, it is evident that the central dot group in the uppermost row, excluding the elongated outliers, shows a strong tendency for smaller Total Floor Area to result in classification as NDTP. Furthermore, being located in areas near a CBD significantly increases the likelihood of being classified as NDTP. Places farther from road geometry also exhibit a strong tendency to be classified as NDTP. Buildings located in commercial zones are more likely to have NDTP. The influence of the SHAP value is significant in both Building Age and Dist. Station, yet the underlying trend remains ambiguous. Conversely, the tendency of the Dist. CBD, Total Floor Area, Dist. Road, and Commercial Zone variables to be categorized as DTP was opposite of that of NDTPS – farther from CBD, not located in commercial zone, and closer to smaller buildings and road geometries. In the case of ADTP, close distance to CBD increases the likelihood of classification as ADTP, with a pronounced effect when lower building age. While building floor area does not exhibit a distinct difference, being more proximate to road geometries increases the likelihood of being classified as ADTP. Notably, proximity to subway stations is associated with a higher likelihood of classification as ADTP.

For the remaining activity categories in Figs. 6 and 7, there are variations in the order of importance and the extent of impact on classification. However, the overall tendencies of the positive and negative effects of each spatial attribute remain consistent across all categories. The SHAP analysis reveals the following key findings: 1) Distance to CBD emerges as the most crucial attribute in distinguishing Third Places by level of digital utility, where being away from CBD has a positive

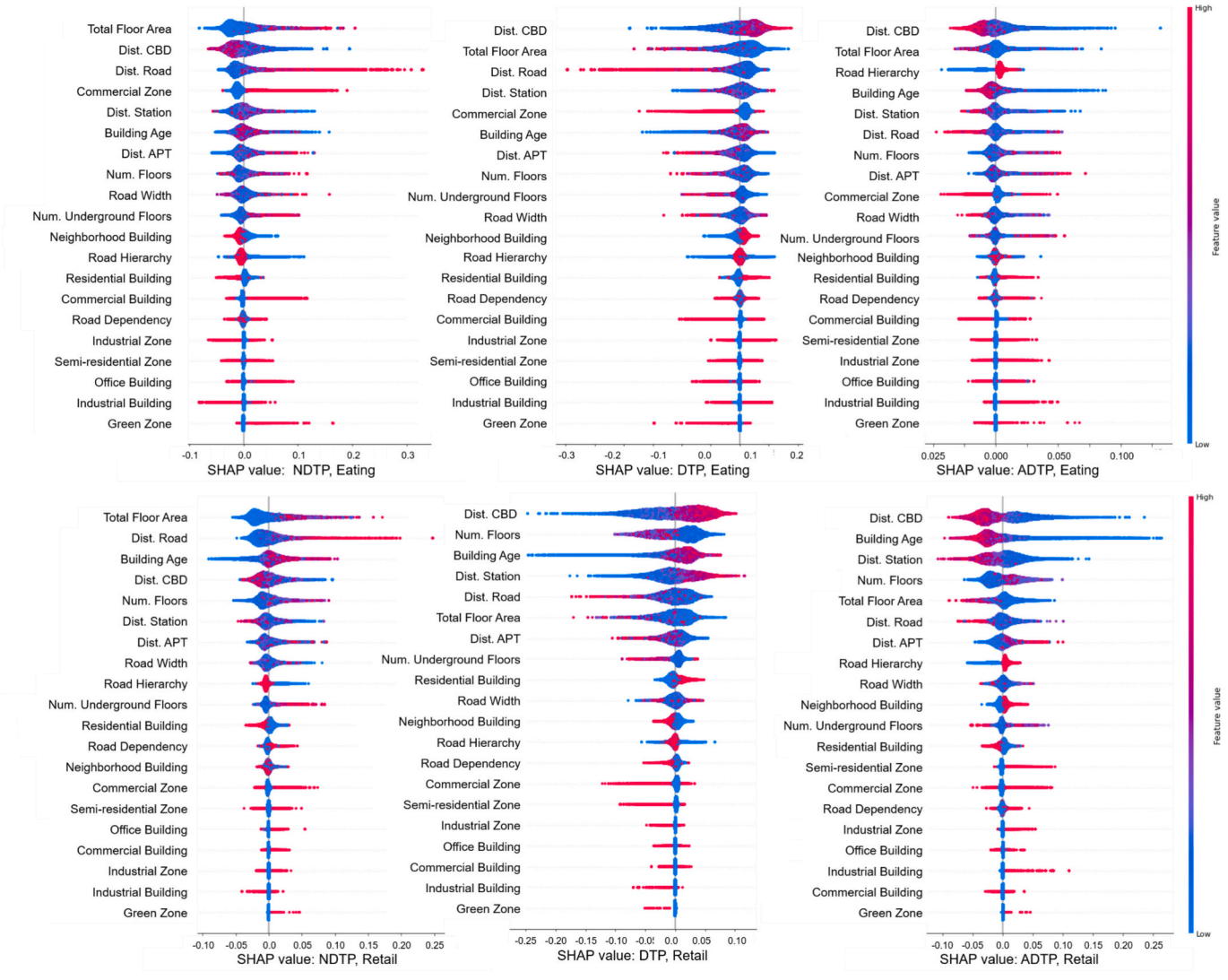


Fig. 6. SHAP value of each spatial attribute of eating and retail places.

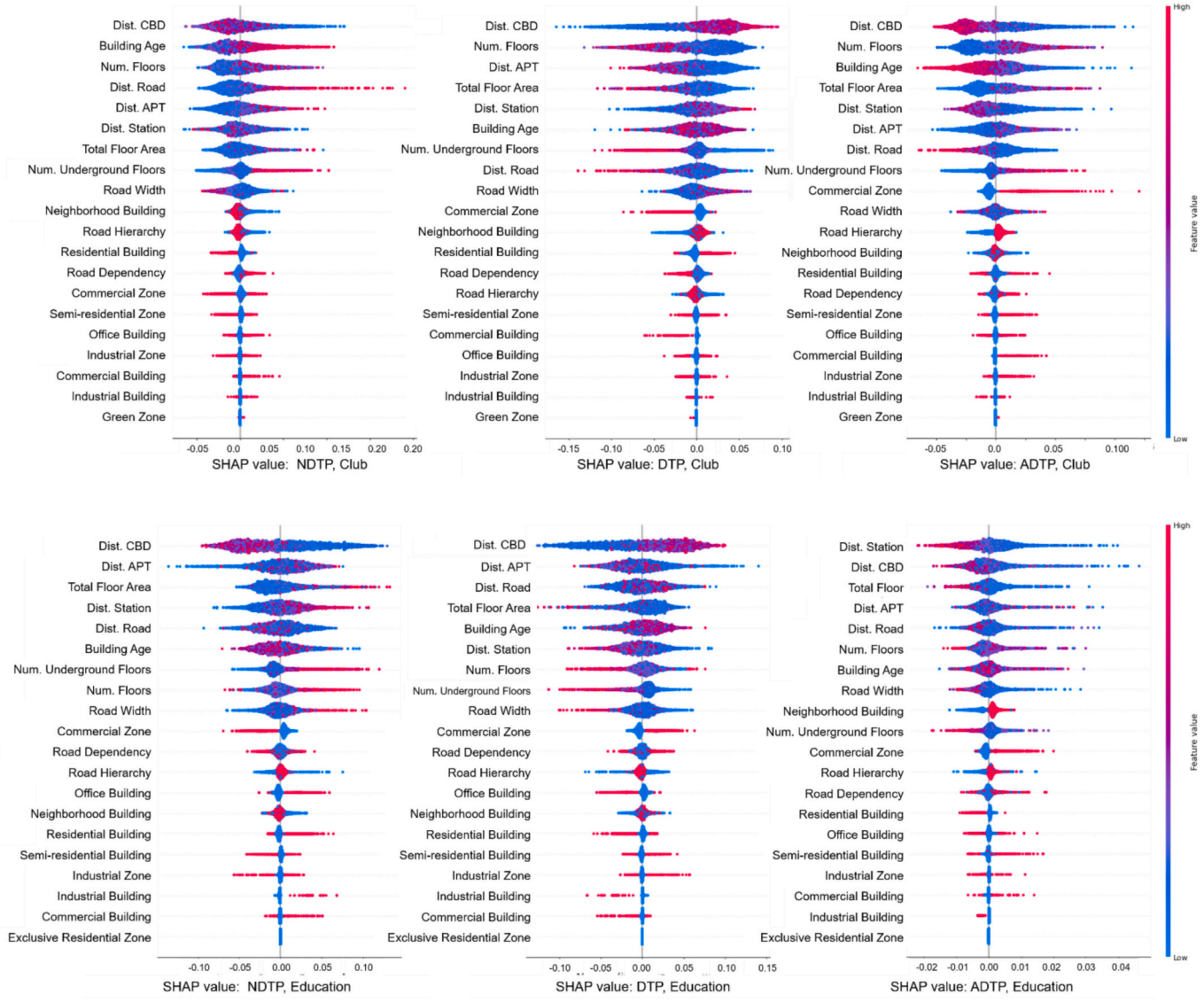


Fig. 7. SHAP value of each spatial attribute of club and education places.

influence on classification as DTP and a negative influence on classification as ADTP; 2) Larger buildings, such as large Total Floor Area and high Num. Floors, are more likely to be classified as NDTPs; 3) NDTP and ADTP exhibit opposing tendencies with respect to Building Age. As Building Age increases, the likelihood of classification as NDTP rather than ADTP also increases. 4) being located within collective buildings, which are characterized by greater distance from road geometry, positively influences classification as NDTP.

4.5. Geospatial analysis and case studies

Following the abovementioned consistent findings in Section 4.4, geospatial analysis shows how the spatial differences between Third Places empirically occur in urban environments. This section presents examples where the properties of quantitatively measured spatial variables are represented in physical space, depending on the type of DTP.

4.5.1. Distance to CBD

Dist. CBD is one of the most influential spatial variables in the classification of Third Places. The order of proximity from the CBD is ADTP-NDTP-DTP, with the latter being the most distant. Table 5 demonstrates that this trend is consistent across entire activity categories. Table 4 exhibits that the gap in spatial variables is statistically significant. Fig. 8 illustrates the locational distribution of the number of NDTPs, DTPs, and ADTPs in a 500 m grid, stratified into five quantiles, and the location of the subway stations (Yeouido, Hongik University, and Gangnam) in the center of Seoul's three CBDs, as well as a 3 km radius around them. As Fig. 8-(c) illustrates, ADTPs are more densely concentrated around the CBD in comparison to NDTP and DTP. A comparison of Fig. 8-(a) and (b) reveals that the distribution of Third places is less discernible, although it can be observed that DTPs are more dispersed throughout the periphery of Seoul, with the exception of the CBD, in comparison to the distribution of NDTPs.

Fig. 9 illustrates the distribution of the Dist. CBD variable according to the activity category, with a representative example provided for reference. Fig. 9-(a), a bar plot of each category's Dist. CBD, depicts that ADTPs are located closer to the CBD than the other two categories of Third Places. Upon examination of the cases based on the actual distribution, we found that ADTPs are predominantly located in densely populated areas in close proximity to major CBDs. For instance, the hair salon situated in the vicinity of Yeouido station, as illustrated in Fig. 9-(c), exemplifies a Third Places that is located in a highly competitive environment due to the concentration of floating populations, which are both potential providers and users. Conversely, in the case of DTPs that

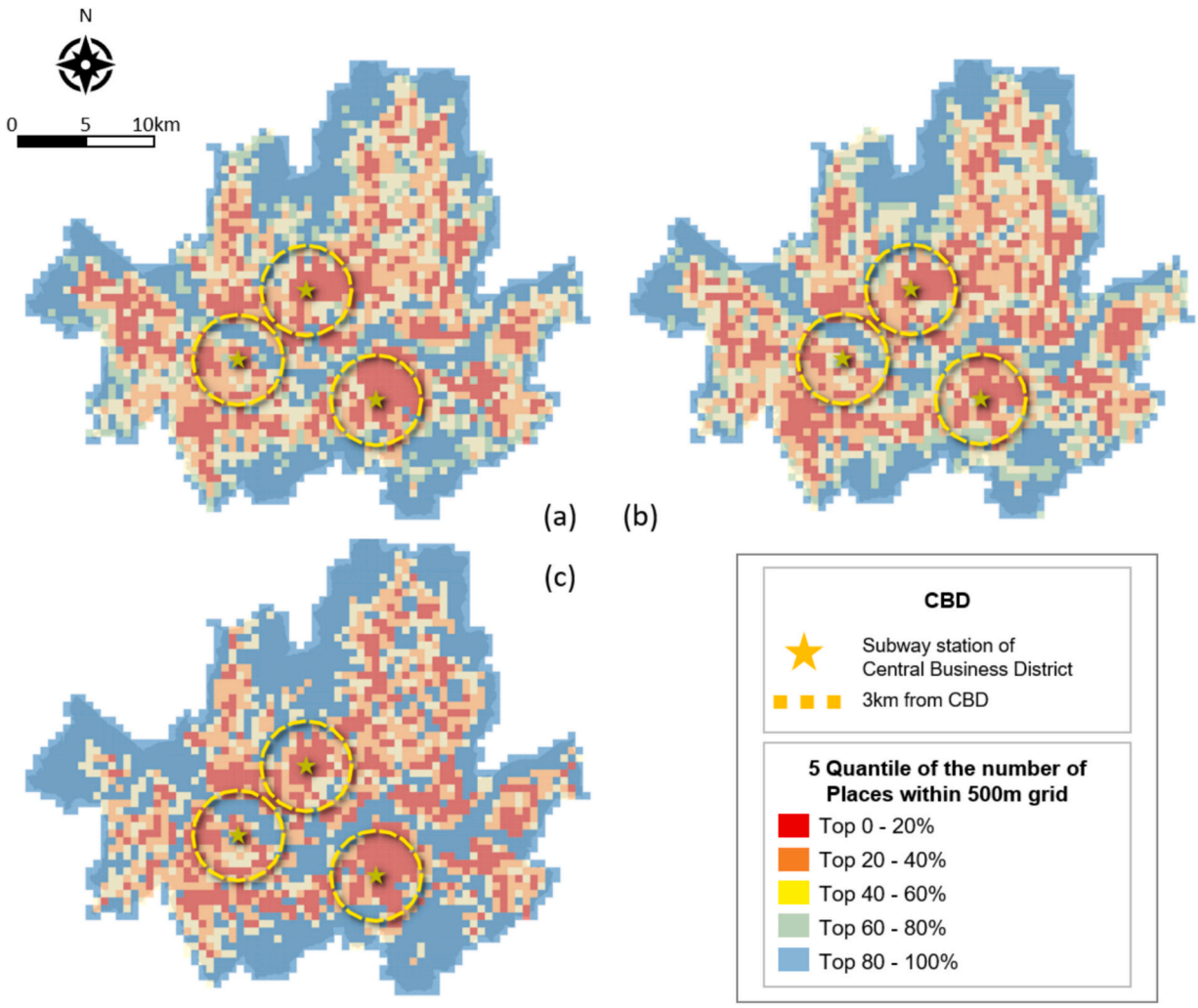


Fig. 8. The location of CBD and the number of Third Places within 500 m grid –(a): NDTP, (b): DTP, and (c): ADTP.

are located at a distance from the CBD, there is a reduced level of competition for participation in social activities. This is exemplified by the hair salon located in a residential area, as illustrated in Fig. 9-(b). However, it is essential to provide additional information, such as location and menu, to enhance the customer engagement.

4.5.2. *Building sizes*

Total Floor Area is a spatial variable that demonstrates the difference in digital utility between Third place located in high-rise complex buildings and small-sized residential-commercial complex buildings. As tabulated in Table 4, there is a statistically significant difference in the activity category. According to Table 5 and Figs. 7 and 8, NDTPs are located in buildings with large Total Floor Area, and it can be observed that the variable tends to decline from DTP to ADTP. Fig. 10-(a) provides a visual representation of the tendency. Through the empirical observation of the spatial database, we identified that a number of NDTPs located in a high-rise complex contributed to the tendency, as exemplified in Fig. 10-(b). The high-rise complex is distinguished by an affluent customer base and a vibrant social environment, which is sufficiently accessible. In contrast, the DTP in Fig. 10-(c), located approximately 200 m from the building in Fig. 10-(b), is located in a small-sized residential-commercial complex. We interpret that the different spatial characteristics of the high-rise complexes and small-sized residential-commercial complex buildings seem to have influenced the Total Floor Area variable.

4.5.3. *Building Age of ADTP*

The influence contribution results from the SHAP analysis highlight that Building Age is the critical variable for classifying Third Places by the level of digital utility. Table 5 indicates that, with the exception of the Education activity category, the building where ADTP is located is relatively recent in construction. This finding is corroborated by Table 4, which demonstrates that this distinction is statistically significant. Fig. 11-(a) presents a comparison of the ages of buildings that encompass Third Places (gray bars), Third Places in each category (yellow bars), and Third Places within each category (the others). Fig. 11-(a) demonstrates that the buildings in which ADTPs are located are markedly more recent in construction. Fig. 11-(b) illustrates the case of a Taekwondo center, a prototypical Third Place where social interaction occurs through sports, where an NDTP is located that is not registered on the NAVER Place. This case can be interpreted as a result of NDTPs, such as taekwondo centers and boxing centers, being located in proximity to residential complexes that were organized a considerable time ago. Conversely, ADTPs are primarily Pilates and fitness centers, located in relatively recently constructed buildings (Fig. 11-(c)). Our observation found that the activities that occur in ADTP predominantly attract users who can utilize reservation service, and the places are located in recently constructed residential and commercial areas.

4.5.4. *Collective buildings*

Two types of buildings significantly influence the extremely large values of Dist. Road: 1) university campuses, which have the longest

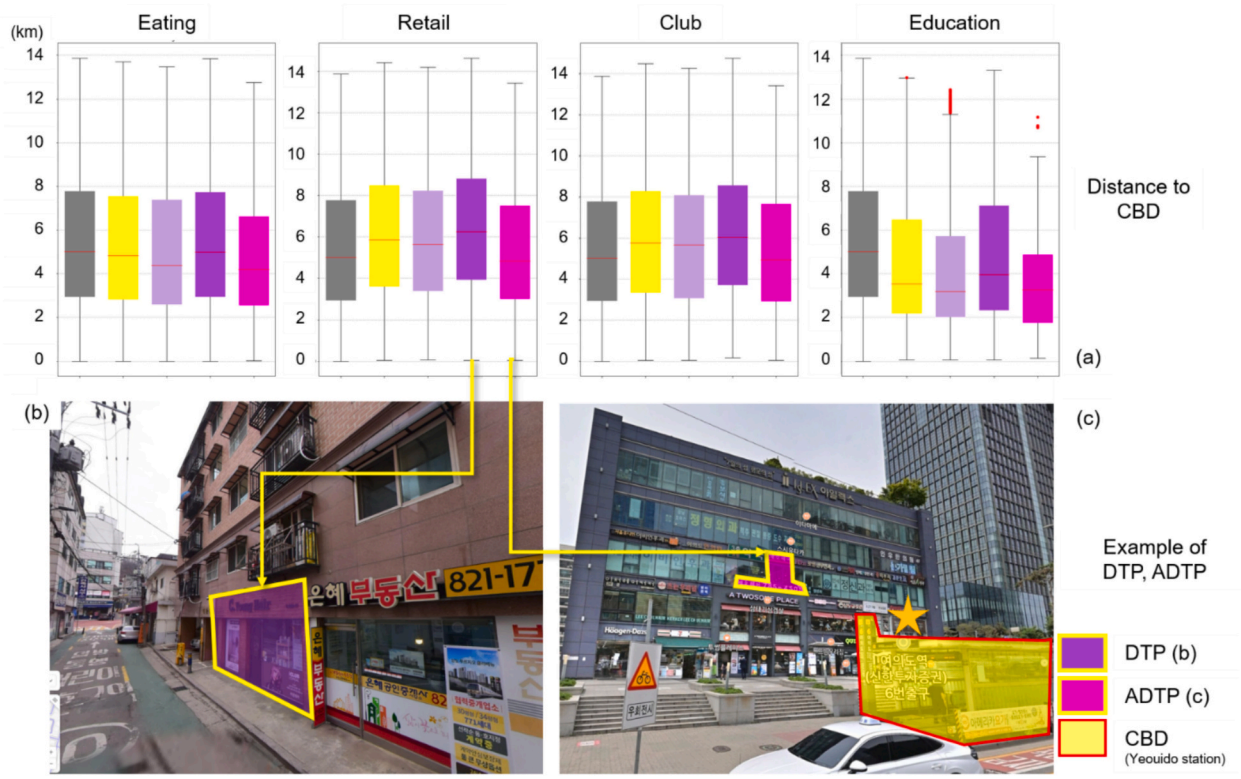


Fig. 9. (a): Bar plot for distance from the CBD of NDTP, DTP and ADTP, (b): an example of a retail type of DTP (c): an example of a retail type of ADTP near by a subway station of CBD (street view images from NAVER maps).

average Dist. Road, and 2) apartment complexes, which constitute the largest proportion of building types >15 m from the road. Both types of buildings share common characteristics in that they are collective buildings. Based on the MOIS data source (Ministry of the Interior and Safety, 2023b), road geometry data within university campuses and apartment complexes are rarely established because of mountainous locations or frequent road repair. As illustrated in Figs. 7 and 8, a larger Dist. Road variable increases the likelihood of being classified as an NDTP in the Eating and Retail activity categories. The statistical significance observed in Tables 4 and 5 corroborates this finding. Fig. 12-(a) visualizes the proportion of NDTPs, DTPs, and ADTPs located within a collective building group (university campus or apartment complex) for each activity category, contrasting the proportion per category with the proportion of the entire dataset. Fig. 12-(a) reveals that the Eating, Retail, and Club category exhibits a notable prevalence of NDTPs located within collective buildings. Fig. 12-(b) shows the example of a bakery located inside a Korea university campus, which has a high Dist. Road variable due to the lack of road geometry data inside the university campus. These NDTPs exemplify the Third Places that facilitate social activities occurring within a geographically confined setting, as depicted in Fig. 12-(c).

5. Discussion

5.1. Distinct spatial characteristics of third places via digital technology

The spatial characteristics of Third Places differ significantly based on the level of digital utility, as demonstrated by the results of the ANOVA F-test in Section 4.2. Most spatial characteristics including location and building dimension show statistically significant differences between classes. This finding demonstrates that Third Place is significantly differentiated in terms of physical environment by the type of platform service it utilizes. The results of the ANOVA F-test provide quantitative evidence that the interaction between physical urban space and digital technology is indeed reshaping urban spaces, in line with the expectations of previous studies (Castells, 2000, 2002; Zook & Graham, 2007a,b). In addition, this study devised a methodology to categorize urban spaces where social activity occurs based on registration and utilization of digital platforms, and compared the unique spatial characteristics of each category. We expect that the methodology and framework presented in this study will contribute to the ongoing academic discussion of technology-induced changes in geography proposed in the field of digital geography and platform urbanism.

5.2. Activity types and digital utility

According to Table 2 the second row of each graph in Fig. 12-(a), the use of digital technologies in Third Places varies based on the types of activities they host. In total, 76.09 % of Third Places were registered on NAVER Place. The high proportion means that most of the third places where social activity occurs are actively utilizing digital platforms. Third Places in Eating category with the highest DTP ratio, which is the largest proportion of the dataset, represent the tendency. The highest ADTP ratio was observed in the Retail category. In places such as hair and nail salons, users plan the time of visit in advance via the platform, which then informs the providers to prepare services based on the information. The utilization on the platform plays a pivotal role in coordinating social activities between providers and users to occur at mutually specified times, a role that is particularly evident in the activities within the Retail category. Club activities exhibit both high NDTP and ADTP rates. As previously mentioned in Section 4.5.3, this difference is due to the contrast between sport clubs, such as taekwondo clubs, which cater to a pre-established user base within existing residential areas, and fitness centers and Pilates classes, which are located in recently constructed areas. Education activity exhibits a markedly high NDTP ratio and a comparatively low ADTP ratio. This might be indicative of the characteristics of private academies, where time slots are predetermined before engaging in. However, interpretation of Education activity requires

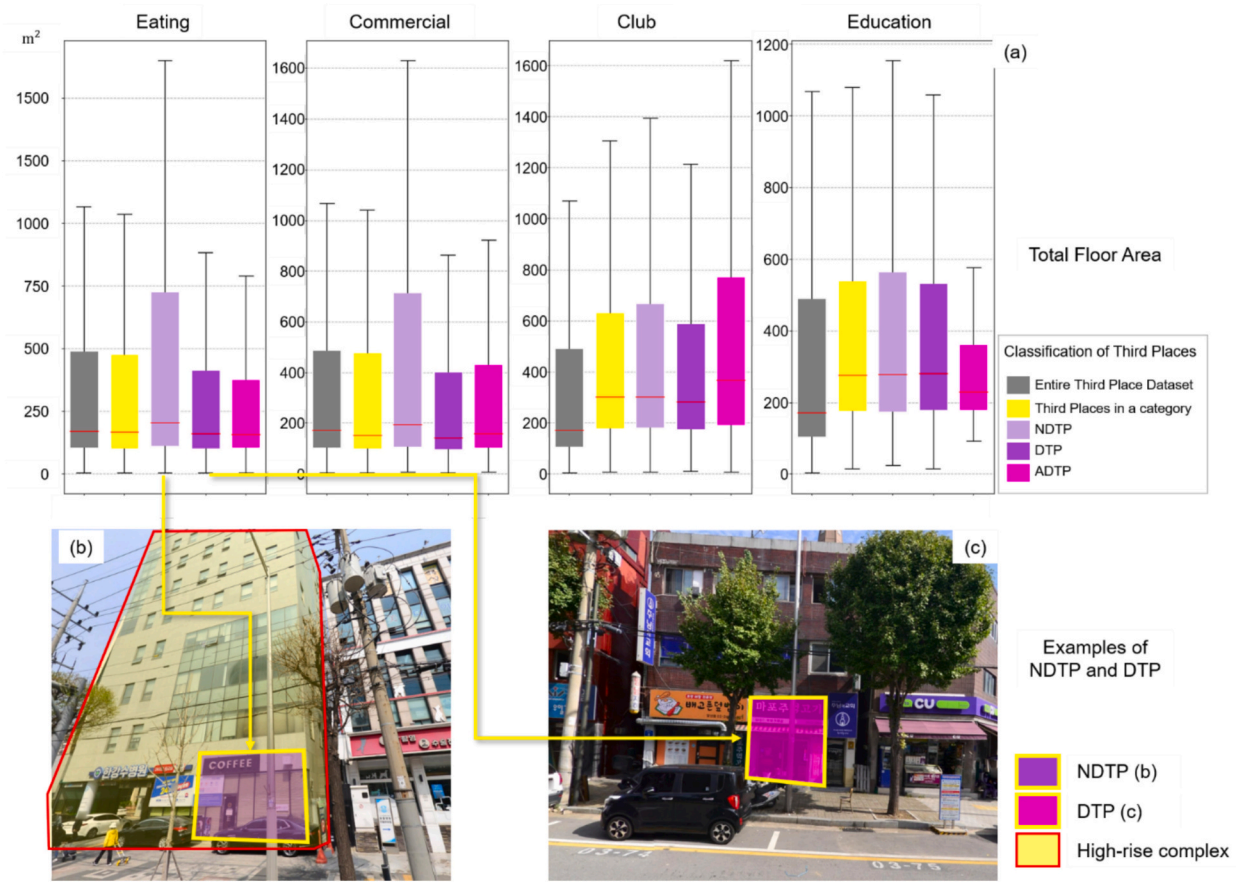


Fig. 10. (a): Bar plot for Total Floor Area of NDTP, DTP and ADTP, (b): an example of an eating type of NDTP in a high-rise complex (c): an example of an eating type of DTP (street view images from NAVER maps).

more clarification due to the relatively small number of data.

In summary, we hypothesize that the findings of this study reveal the empirical influence of digital technology on social activities in urban spaces, which are mediated by platforms, in three ways: 1) accessibility to a place (the provision of information and expansion of choice, as in the example of a small-sized residential-commercial complex); 2) time scheduling (the reservation before visiting a hair salon); and 3) transformation of mode to participate in activities (taekwondo club for pre-organized residence and Pilates class for recently developed areas). This finding is consistent with existing studies indicating that formal activities, including those related to living and working, will be influenced by changes in the residence and working places due to digital technologies (Kwan et al., 2007; Mokhtarian et al., 2006). However, because of the involvement of the providers and users of services, which creates a faster cycle of change, the informal activities exhibit different characteristics from formal activities. Therefore, further comparative studies are suggested to reveal the impact of digital technology on a wider range of activities.

5.3. DTP: augmented accessibility

The spatial characteristics of DTPs indicate that digital technology offers a novel form of accessibility that is distinct from the physical accessibility of Third Places. Fig. 10-(c) depicts an example of a restaurant located within a small-sized residential-complex complex. In comparison to the case in Fig. 10-(b), the Third Place illustrated in Fig. 10-(c) has relatively low physical accessibility. Table 5 tabulates that a higher digital utility is associated with a higher hierarchy of the nearest roads (closer to lower-hierarchy roads) for all activity categories and a small width of the nearest roads, particularly within the Eating and Retail category. The SHAP analysis in Figs. 6 and 7 indicates that the higher the road hierarchy, the higher the likelihood of being classified as ADTP. This suggests that Third Places located on less accessible roads or in alleys are more likely to utilize digital technology.

These findings align with the results of previous studies that used qualitative methods to address the impact of ICT on human travel (Hong et al., 2020; Mokhtarian et al., 2006). They suggested that people plan trips with the aid of additional information before visiting places. The observation that DTP locations are farther from subway stations than NDTP locations in the Eating and Retail activity categories supports this idea. In contrast to traditional methods of acquiring information about a place, such as visual exposure or word-of-mouth recommendations, Third Places with a high level of digital utility, which provide and showcase information through digital technology, have the flexibility to choose less physically accessible locations (Park & Kim, 2024). This suggests that digital technology facilitates people exploring and visiting less accessible urban spaces to participate in social activities offered by Third Places.

5.4. NDTP: advantaged accessibility to sufficient visitors

Third Places that have already secured advantaged accessibility to sufficient visitors do not need to use digital utilities. Two findings support this interpretation.

First, Third Places with affluent customers tend to rely less on platforms based on digital technology. As demonstrated in Table 5, Figs. 6 and 7, the buildings where NDTP is located exhibit the following characteristics: proximity to wide, high hierarchy (accessible) roads, large Total Floor Area (high-rise complex). Fig. 13-(a) exhibits the exemplary place that secured sufficient visitors by catering to the working

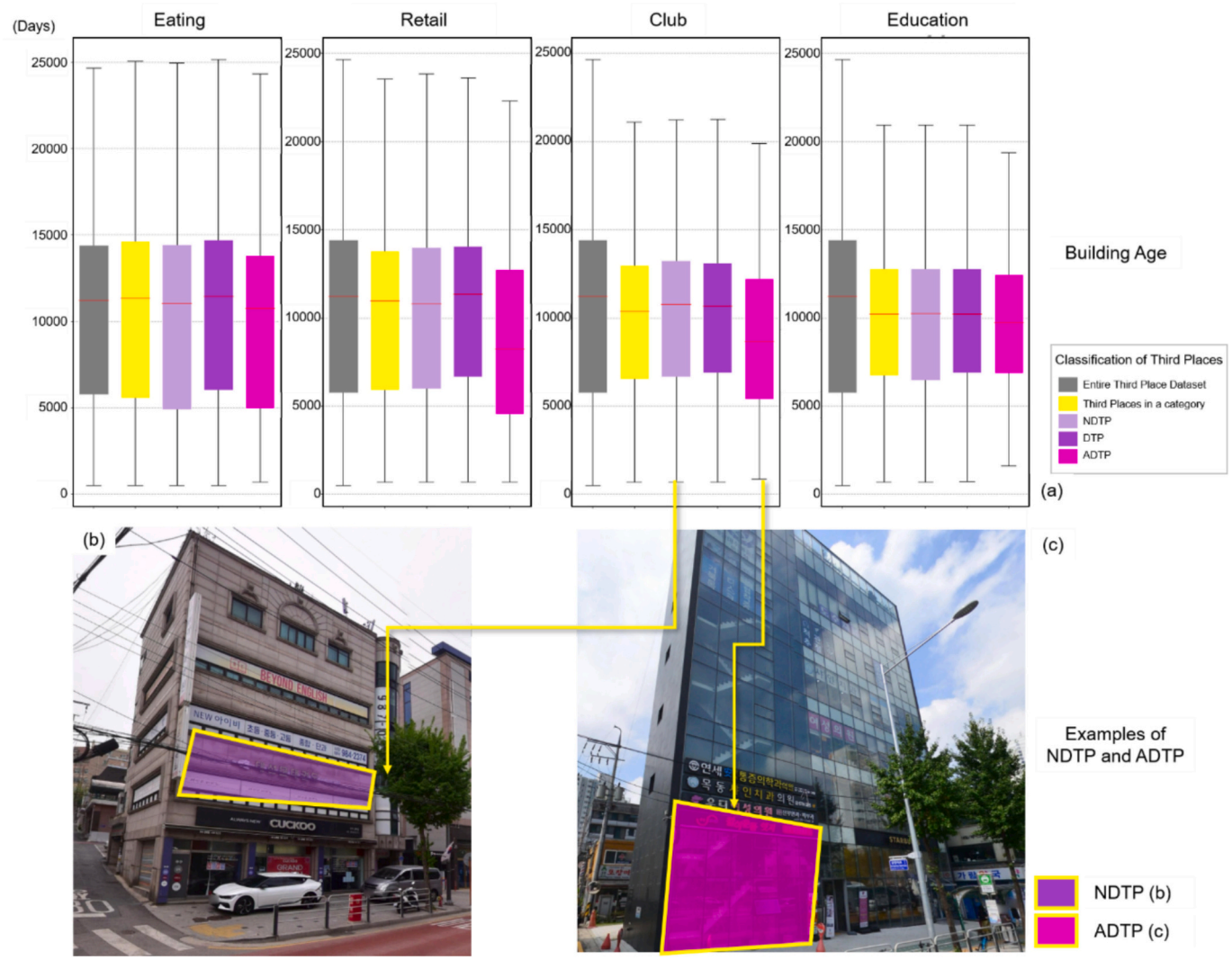


Fig. 11. (a): Bar plot for Building Age of NDTP, DTP and ADTP, (b): an example of a club type of NDTP in an old building (c): an example of a club type of ADTP in a new building (street view images from NAVER maps).

population in the high-rise office building. The high-rise complex located in conveniently accessible environment and commercial zones, which attract a substantial visitors base from the working population and tourists, may not require an online platform for provide and participate in social activities.

Second, Third Places located at a considerable distance from the road geometry, which suggests that they are within collective buildings, have the advantages of a local community of patrons and residents. Fig. 13-(b) is an example of a Third Place that is located within a community center nearby residential neighborhoods, sufficiently accessible from the local community. The consistent observation that the Dist. Road attribute for NDTP is larger than that for DTP and ADTP indicates that NDTP locations are more likely to be on university campuses or in apartment complexes. The SHAP analysis underscores the significance of the Dist. Road attribute in classifying NDTP, DTP and ADTP.

These results indicate that the utilization of digital platforms is less probable in Third Places located in spaces that have advanced accessibility for sufficient visitors. Businesses located in collective buildings already secure sufficient potential patrons to operate effectively without the need for extensive digital technology. These findings align with previous research, such as Hatuka et al. (2021), which has suggested that local residents who routinely visit local retail shops are less likely to use ICT for their shopping activities. Therefore, Third Places located in areas with advantaged accessibility to sufficient visitors or a strong local context may not find it necessary to rely on digital technologies.

5.5. *ADTP: service competition aid*

The spatial characteristics of ADTP reveal that people utilize advanced digital technologies to secure priority in severe service competition. According to Table 5, ADTP exhibits significantly lower values in Dist. CBD, Dist. Station and Building Age than NDTP and DTP. First, Third Places in locations where social activity can occur competitively can be interpreted as utilizing the online platforms' reservation services. The CBD and station influence area are the most profitable districts with the proliferation of floating populations owing to fixed commuting demands and public transit (Tu et al., 2020). As shown in Fig. 9-(c), this suggests that Third Places in areas with affluent floating populations and well-established commercial resources provide online reservation services. Rather than seeking service in more distant locations, people opt for online reservations to secure services in areas with fierce competition and limited available spaces (Park & Kim, 2024).

Second, the low age of buildings associated with ADTP indicates that newly constructed or recently redeveloped commercial spaces are more likely to adopt online reservation services. Figs. 6, 7, and 11(b) support this finding. The figures suggest that Third Places with advanced digital technologies are persistently opened in newly constructed buildings, pointing toward a future trend of such technologies continuing to be



Fig. 12. (a): Proportion of Third Places within collective buildings, (b): Korea university campus and NDTP within the campus (c): an example of NDTP in the university campus.

integrated into upcoming developments.

6. Conclusion

This study provides a quantitative analysis of whether Third Places exhibit varying spatial characteristics based on the level of digital utility and, if so, which attributes influence the differences. To do so, we define the "Digi-Third Places" concept by categorizing Third Places into three distinct classes according to the level of digital utility required to participate in social activities. Through three quantitative approaches, we determine the spatial differences of Third Places via the level of digital utility. Our empirical findings underscore the significant relationship between digital technology and the spatial differentiation of Third Places within physical environments. This research framework offers a valuable contribution by facilitating empirical analyses.

While this study suggests a novel approach to quantitative analysis in the digital geography field, it has several limitations. First, the classification of NDTP, DTP, and ADTP, as well as the types of activities, has certain limitations. Generalizability and addressing the data imbalance caused by the scarcity of ADTP data demand further research based on a comprehensive dataset. Second, the findings of this study are context-dependent and focus primarily on Seoul. Although Seoul is an adequate digital geography research site due to its high digital affinity, broader generalizability would necessitate further investigation in various international settings. Third, the scope of digital integration covered in this study is limited to the use of platforms by service providers and users, and the spatial features of the places that utilize the technologies. To consider the complex nature of the interaction between digital technology and urban space, i.e. the broader spectrum of digital interaction, it is necessary to examine and analyze a wider range of

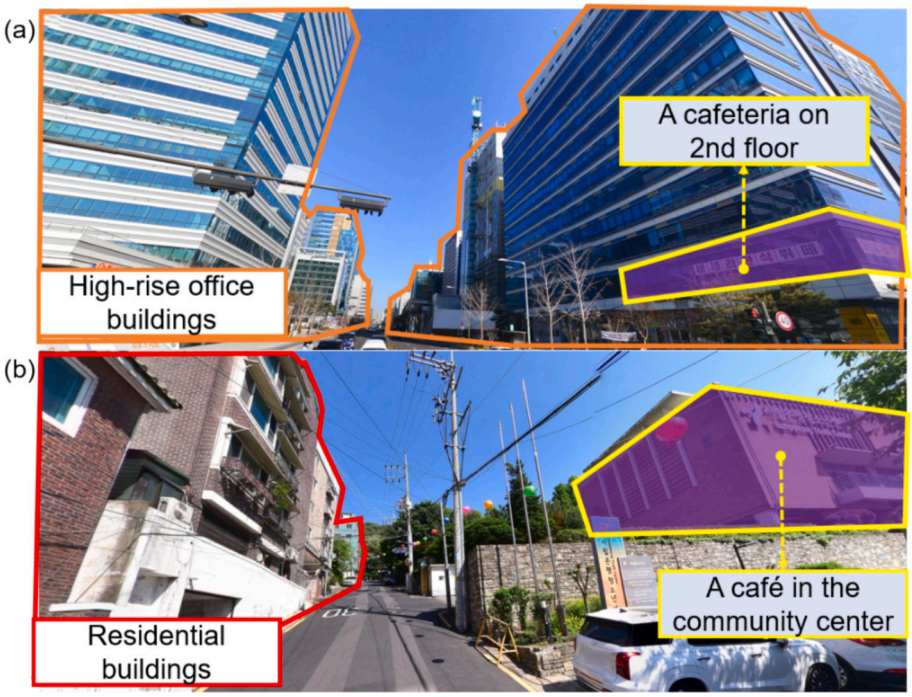


Fig. 13. The examples of NDTP that provide catering services for visitors in the high-rise buildings - (a): a cafeteria of a high-rise business building in Gasan, (b) a café located in a local youth community center (Eunpyeong Youth Center).

digitally-mediated activities, including the investigation on user experiences.

We suggest further study of the social interaction and human behavior induced by digital technology. First of all, temporal analysis utilizing time-series data will further support the implications of this study. Although this study could not be conducted due to lack of data, temporal analysis is expected to provide important clues to track the impact of digital technologies and platforms on urban space and the dynamics of changes in the urban environment. Moreover, further comparative studies conducted using an expanded dataset from international platforms and global cities would contribute to the exploration of the study's results in Seoul within a worldwide context. Second, our findings shed light on the broader discourse on the digital divide, which emphasizes how varying levels of affinity with digital technologies can influence the distribution of opportunities and social resources (Scheerder et al., 2017; Van Dijk, 2006; Van Dijk & Hacker, 2003). Additionally, we look forward to further research that instruments a critical view to examine the rapidly evolving digital technologies that are transforming urban spaces, such as the inequality of data and information caused by corporate-led platforms (Graham & Dittus, 2022; Zook & Graham, 2007a,b) and the privatization and fragmentation caused by algorithms (Habermas, 2022; Thiel, 2023).

CRediT authorship contribution statement

Chulwoong Park: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Youngchul Kim:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Role of the funding source

The funder had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

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Declaration of competing interest

No potential conflict of interest was reported by the authors.

Data availability

Data will be made available on request.

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